

No Legal Claim Beyond Its Carrier: Claim-Status Warrant under AASC

Evidentiary Carrier Adequacy, Local Promotion Control, and Report-Preserving Legal Inference

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Abstract

Legal systems distinguish relevance, authentication, admissibility, weight, burden, expert qualification, chain of custody, and permissible use. A recurrent failure cuts across those categories: a claim is reported at a legal or evidentiary status stronger than its carrier licenses. Authentication becomes substantive truth; admissibility becomes proof of the asserted legal target; expert credentials become method reliability; similarity becomes source identity; machine output becomes evidentiary warrant; witness agreement becomes independent corroboration; and narrative coherence becomes causation. These are not merely ordinary evidentiary errors. They are illicit claim-status promotions.

This paper develops the legal-domain instantiation of the Admissibility And Standing Constraint (AASC) / Universal Exactness Audit Protocol (UEAP) architecture. The central object is claim-status warrant: permission, under a declared jurisdictional and procedural ledger, to report a legal or evidentiary claim at a requested status class. Evidence law supplies the domain standards; AASC supplies the cross-status audit grammar that prevents reports from exceeding their target, carrier, alternative-disposition, laundering, and report-preservation conditions.

The paper gives a doctrine-facing warrant architecture rather than a replacement for legal doctrine. It develops legal proof as status-preserving construction: evidence, narratives, probabilities, explanations, expert opinions, and machine outputs may operate as carriers, but a legal proof-report is warranted only when the transition from evidence to report preserves target, carrier, burden, alternative disposition, factfinder allocation, and report limits. The paper also develops a status-vector model, a local-promotion graph, an alternative-disposition account adapted to burdens and factfinder roles, a legal laundering taxonomy, evidence-network rules, a registry protocol, and a court-facing checklist. Worked audits cover cartridge-case source attribution, expert overstatement, authentication, causation, and AI/machine-generated evidence. The practical rule is simple: no legal claim should be reported at a strength greater than its carrier permits.

Keywords: AASC; UEAP; claim-status warrant; evidence law; expert testimony; forensic science; admissibility; probative value; Rule 702; Daubert; Mohan; White Burgess; chain of custody; machine-generated evidence; AI evidence; laundering exclusion; reportability; legal inference.

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1 Introduction: legal status promotion as the hidden failure mode

Law already contains a dense vocabulary for controlling evidence and inference: relevance, admissibility, probative force, unfair prejudice, authentication, chain of custody, expert qualification, method reliability, burden of proof, standard of review, and permissible use. Yet many important legal failures do not occur wholly inside one of those categories. They occur when a claim is moved from one category to another without carrying the conditions required for that stronger category.

The recurring pattern is status promotion. Similarity is reported as identification. Admissibility is reported as proof. Expert authority is reported as method validity. Method validity in general is reported as reliable application in a case. Narrative coherence is reported as causation. A likelihood relation is reported as categorical source certainty. Authentication is reported as substantive truth. Multiple agreeing witnesses are reported as independent convergence without a contamination audit. A machine-generated output is reported as evidentiary warrant. A civil-strength inference is reported in language that sounds criminal-strength. These failures are not reducible to falsity. Some promoted claims may be based on genuine observations, real documents, competent experts, useful models, or admissible evidence. The problem is that the status reported to the legal user exceeds the warrant-bearing carrier.

The legal problem addressed here is therefore not merely whether a piece of evidence is true, relevant, admissible, persuasive, or useful. The prior question is what legal status the asserted claim is entitled to carry. The practical surface of the paper is a court-facing checklist: what is the claim, what status is requested, what carrier supports that status, what alternatives remain, what local promotion edge is being crossed, and what wording would overstate the carrier? The Admissibility And Standing Constraint (AASC) does not demote expert testimony, forensic science, machine evidence, or institutional expertise. It protects their proper legal status by preventing over-reporting: a validated method can be warrant-bearing, an expert report can be probative, a machine output can be useful, and a forensic comparison can be powerful when the carrier and report language match the status asserted. A legal claim may carry warrant only as authenticated. It may be admissible only for a limited purpose. It may be evidence-compatible without being source-identifying. It may be method-valid within a declared domain without being reliable in application to this case. It may be causally suggestive without satisfying a jurisdiction's causation standard. It may be a machine computation without yet being legal evidence.

This paper develops a legal-domain instantiation of the AASC/UEAP protocol developed in Maley (2026a). UEAP treats legitimacy as status-relative: a claim is legitimate at a requested status only if target fixation, carrier adequacy, alternative disposition modulo skin, laundering exclusion, and report preservation all hold. The legal paper imports that architecture into law while leaving jurisdiction-specific doctrine in place. Evidence law supplies the relevant standards. AASC supplies the audit grammar that prevents compatibility, authority, computation, narrative coherence, and admissibility from being laundered into proof, causation, source identity, or legal actionability. For philosophy of law and legal epistemology, this is the paper's proof-theoretic claim: legal proof is not merely a high probability, a coherent narrative, an admissible record, or a best explanation; it is a status-preserving construction whose final report must remain inside the carrier that warrants it.

1.1 Core thesis

Let J denote a legal audit ledger: jurisdiction, procedural posture, evidentiary rules, burden standard, permitted inference type, factfinder role, expert-testimony standard, authentication standard, exclusionary constraints, and report boundary. Let c be a legal or evidentiary claim, and let σ be the requested legal status class. The central certification condition is:

$$\begin{aligned} & \text{ClaimStatusWarrant}_J^\sigma(c) \\ & \iff \\ & \text{TargetFix}_J(c) \wedge \text{CarrierAdeq}_J^\sigma(c) \wedge \\ & \text{AltDisposition}_J^\sigma(\text{AdmAlt}_J(c) / \sim_{\text{Skin}}) \wedge \\ & \text{NoLaunder}_J(c) \wedge \text{ReportPreserve}_J^\sigma(c). \end{aligned} \tag{1}$$

This biconditional is internal to a declared legal audit basis. It does not assert that all law, all institutional legitimacy, or all legal interpretation reduces to AASC. It says that, once a jurisdictional ledger, requested status, carrier standard, admissible-alternative relation, skin relation, laundering taxonomy, report rule, and status-vector structure are declared, a legal claim cannot responsibly be reported at status σ unless the five audit conditions are jointly satisfied.

Definition 1.1 (Claim-status warrant). *“Warrant” means permission, under the declared audit basis, to report a claim at a requested legal or evidentiary status class. It does not mean constitutional warrant, search warrant, moral warrant, judicial ruling, admissibility ruling, ultimate legal entitlement, or doctrinal standing to sue. Claim-status warrant is report-relative: it concerns what the carrier licenses the speaker, expert, court-facing report, registry, or downstream user to say without changing the audited claim.*

1.2 Contributions

The paper contributes eight things.

First, it develops a target/carrier account of claim-status warrant. A legal assertion must have a fixed target and a warrant-bearing carrier before admissibility, probative force, source identification, causation, actionability, element-sufficiency, or an outcome relation can be responsibly assigned.

Second, it replaces a single taxonomy with a status-vector and local-promotion model. Legal statuses are not points on one ladder; the audit tests local promotions such as authentication to truth, admissibility to proof, general method validity to case-specific application, similarity to source identity, and machine output to warrant for the asserted legal target.

Third, it formalizes legal laundering. Legal laundering occurs when a transformation changes the apparent legal status of a claim without discharging the blocker separating the weaker status from the stronger one.

Fourth, it develops a proof-theoretic account of legal warrant. A legal proof-report is treated as a status-preserving construction from target to report under a jurisdictional ledger, not merely as a probability, narrative, explanation, admissible record, or expert assertion.

Fifth, it gives a protocol-result spine for legal status preservation: target fixation necessity, carrier adequacy, freedom exhaustion, laundering exclusion, evidence-network non-amplification, similarity/source-identity separation, machine-output non-warrant, burden-laundering exclusion, and report preservation.

Sixth, it produces an operational registry protocol. Each audited legal claim receives a row recording target, carrier, jurisdiction, status class, evidence/provenance, method, alternatives, network dependencies, laundering risks, report language, final status, blockers, and next action.

Seventh, it gives a detailed forensic centerpiece: a cartridge-case source-attribution audit at operational resolution, including class characteristics, individual characteristics, subclass characteristics, consecutive-manufacture concerns, examiner discretion, contextual information, blind verification, error-rate evidence, inconclusive-category handling, and report-language discipline.

Eighth, it develops machine-generated evidence as the pure legal form of computation laundering and gives a separate system/data/validation/error/domain/report ledger for AI or machine outputs.

1.3 Nonclaims and boundary conditions

Nonclaim 1.1 (Limits of the legal AASC audit). *This paper does not provide legal advice, decide admissibility, determine guilt or liability, replace jurisdiction-specific doctrine, replace judicial gatekeeping, replace the jury or factfinder, override statutory or constitutional law, or certify any real-world litigation position. It supplies a status-preserving audit protocol for legal and evidentiary claims.*

The paper also does not claim that every legal system uses the same evidentiary carriers. Quite the opposite: different jurisdictions, forums, procedures, and burdens may supply different carrier standards. AASC does not determine those standards. It requires them to be declared before a claim is promoted to the requested status.

This distinction is essential. In the United States, Rule 702 and the Daubert/Kumho line supply important expert-testimony gatekeeping standards (United States Courts, 2024; Fed. R. Evid. 702, 2026; *Daubert*, 1993; *Kumho Tire*, 1999). In Canada, Mohan and White Burgess supply important expert-evidence admissibility and independence anchors (*Mohan*, 1994; *White Burgess*, 2015). Forensic-science reform sources such as the National Academies report and PCAST report identify method-validity, error-rate, and feature-comparison concerns (National Research Council, 2009; President’s Council of Advisors on Science and Technology, 2016). NIST materials and OSAC standards supply further domain-standard resources (National Institute of Standards and Technology, 2026, 2023). These are not replaced by AASC. They enter the audit through the jurisdictional ledger and carrier standard.

The doctrinal illustrations in this legal-domain version are drawn primarily from U.S. federal evidence law and Canadian expert-evidence doctrine because those sources provide clear public formulations of target, carrier, method, application, and report boundaries. The protocol is not limited to those systems. Any civil-law, administrative, arbitral, regulatory, international, or specialized tribunal use must first supply its own jurisdictional ledger before claim-status warrant can be assigned.

2 Claim-status warrant and AASC/UEAP bridge

2.1 The kernel-derived bridge

The legal audit is not introduced as an arbitrary checklist. It is the legal-domain surface of four AASC roles. The source UEAP manuscript separates reference, standing, admissibility, and irreversibility

as kernel roles and translates them into claim-audit roles: target fixation, carrier warrant, admissible continuation or alternative exhaustion, and no laundering/no same-act repair (Maley, 2026a,b). In the legal setting, the translation is direct.

Table 1: Kernel-to-legal bridge.

AASC role	UEAP role	Legal role
Reference	Target fixation	Fix the legal/evidentiary object: relevance, authentication, admissibility, source identity, causation, liability, burden satisfaction, or permitted inference.
Standing	Carrier adequacy	Identify the warrant-bearing legal carrier: testimony, document, chain ledger, forensic method, expert report, statute, precedent, machine ledger, or probability model.
Admissibility	Alternative disposition and licensed continuation	Close, classify, or preserve admissible rival hypotheses under the requested status.
Irreversibility	No laundering and no same-act repair	Prevent a later transformation, report, expert gloss, computation, or institutional label from upgrading an unresolved blocker as if it had been discharged.

The present paper does not rederive the full AASC kernel. It imports the kernel only at the level required for claim-status warrant. The legal audit results below are conditional protocol results: once the legal audit basis is declared, the status assignments follow from the defined audit objects and precedence rule.

2.2 Why legal doctrine alone does not remove the need for the audit

Legal doctrine already contains many gates, but those gates are local. A court may ask whether evidence is relevant, whether it is authenticated, whether a method is reliable, whether an expert is qualified, whether the probative value is substantially outweighed by listed dangers, whether an item falls within a hearsay exception, whether a record was preserved, or whether a burden has been met. AASC asks a cross-status question: what claim, at what requested status, is the proponent trying to report from this carrier?

A claim may satisfy one legal gate while failing another status transition. Evidence may be admissible but weak. An expert may be qualified but the method may lack adequate validation. A method may be generally reliable but misapplied to the facts. A likelihood ratio may be valid under a declared population but not support categorical source identity. A document may be authenticated but not prove its asserted substantive content. AI output may be generated and preserved but not

validated. These distinctions are routine legal concerns, but they become more controllable when represented as target, carrier, freedom, laundering, and report fields.

2.3 Proof-status architecture

The manuscript uses three argument layers.

1. **Definitions** fix legal audit objects: legal target, legal carrier, jurisdiction ledger, requested status, admissible alternatives, skin, laundering route, report object, blocker, and certificate.
2. **Protocol results** derive legal status consequences from those objects under a declared audit basis.
3. **Kernel rationale** explains why the protocol tracks target, carrier, admissibility, and irreversibility rather than some unrelated list.

This architecture prevents two misreadings. The legal paper is not merely a list of practical warnings. Nor does it claim to replace the legal system with a foundational AASC theory. It is a legal-domain audit protocol whose results are conditional on the declared legal basis.

2.4 Proof-status warning: protocol results, not freestanding legal propositions

The formal statements below are intentionally labeled as protocol results, legal audit lemmas, or doctrinal interface propositions rather than freestanding legal propositions. Their force is conditional and internal. Once the target, carrier standard, admissible alternatives, laundering taxonomy, report rule, and status-vector structure are fixed, the validator has no licensed route to a stronger report outside those coordinates. A reader may reject the AASC kernel or alter a jurisdictional carrier standard; that changes the audit basis. It does not license promotion inside an already-declared basis.

This distinction is essential for legal audiences. The paper does not claim that a court is mathematically forced to decide admissibility, weight, guilt, liability, or remedy by AASC. It claims that if a proponent reports a claim at a strength exceeding its target, carrier, alternative-disposition, laundering, and report fields, the report has changed the claim-status object being offered. That is the narrower protocol-level result.

3 Legal audit basis, status vectors, and AltDisposition

Definition 3.1 (Legal audit basis). *A legal audit basis is a tuple*

$$\mathcal{A}_J = (J, \sigma, C_J^\sigma, \text{AdmAlt}_J, \sim_{\text{Skin}}, L_J, \preceq_J^\sigma, R_J),$$

where J is the jurisdictional and procedural ledger; σ is the requested legal status; C_J^σ is the carrier-standard family for that status; AdmAlt_J is the admissible-alternative relation; $\sim_{\text{Skin}}$ is the legally harmless variation relation; L_J is the legal laundering taxonomy; $\preceq_J^\sigma$ is the declared status-precedence order; and R_J is the report-preservation rule.

Definition 3.2 (Legal alternative disposition). $\text{AltDisposition}_J^\sigma(\text{AdmAlt}_J(c) / \sim_{\text{Skin}})$ records what the legal audit does with rival hypotheses, alternative explanations, alternative legal characterizations, or alternative uses of the evidence. In law, alternatives need not always be scientifically exhausted.

They may be excluded by rule, rendered immaterial, defeated under the declared burden, preserved for the factfinder, left open with report-limiting effect, or fatal to the requested status.

Table 2: Alternative-disposition outcomes.

Alternative disposition	Meaning	Report consequence
Legally excluded	The alternative is unavailable in the posture because a rule, privilege, constitutional constraint, sanction, or procedural bar removes it from permitted use.	Does not block the requested report unless the exclusion itself is contested or the report relies on excluded matter.
Immaterial	The alternative remains possible but does not affect the declared legal target, element, status, or permitted use.	No report downgrade, provided immateriality is declared.
Burden-defeated	The alternative remains logically possible but is insufficient under the declared threshold or burden.	Report may proceed only in burden-sensitive language.
Factfinder-preserved	The alternative remains for weight, credibility, or ultimate inference assigned to the factfinder.	Report must preserve the factfinder role and avoid verdict-like language.
Report-limiting	The alternative blocks stronger wording but not all evidentiary use.	Weaker report only: e.g., compatible, source-consistent, or probative-with-limits.
Fatal blocker	The alternative prevents the requested warrant because it remains status-changing under the declared basis.	No requested-status warrant until discharged or reclassified.

The audit basis must be declared before the claim is classified. If the requested status is “admissible expert opinion in federal civil litigation,” Rule 702 and related doctrine enter the carrier ledger. If the requested status is “properly qualified expert evidence in Canada,” Mohan and White Burgess type criteria enter the carrier ledger. If the requested status is “forensic source identification,” method validation, comparison class, error-rate discipline, chain of custody, and report language enter the carrier ledger. If the requested status is “machine-generated evidence without a testifying expert,” current and proposed rules for such evidence enter the ledger as jurisdictional standards rather than as AASC commands (Advisory Committee on Evidence Rules, 2026; Raymond, 2026).

3.1 Jurisdiction and procedure ledger

The jurisdictional ledger records at least the following fields:

Table 3: Minimum jurisdiction/procedure ledger.

Ledger field	Audit content
Jurisdiction	Legal system, court, forum, tribunal, agency, or arbitral body.
Procedural posture	Trial, pretrial motion, admissibility hearing, summary judgment, sentencing, administrative proceeding, appeal, investigation, or internal review.
Case type	Criminal, civil, administrative, regulatory, family, immigration, commercial, or hybrid.
Requested status	Relevance, authentication, admissibility, expert opinion, probative weight, source identity, causation, liability, guilt, remedy, actionability, element-sufficiency, or an outcome relation.
Burden or threshold	Probable cause, prima facie showing, balance/preponderance, clear and convincing, beyond reasonable doubt, or jurisdiction-specific standard.
Evidence rule	Relevance, authentication, hearsay, expert testimony, character, privilege, prejudice/probative balancing, or other rule family.
Factfinder role	Judge, jury, agency, tribunal, arbitrator, administrative officer, or mixed body.
Permitted use	Limited purpose, full substantive use, impeachment, background, notice, state of mind, expert basis, demonstrative support, or no permissible use.
Report boundary	Exact wording that may be used without changing status.

3.2 Requested legal status classes

A legal claim is always status-indexed. The same evidence may satisfy a low threshold while failing a high threshold. The audit therefore uses requested status classes rather than a single binary label.

Definition 3.3 (Legal status class). *A legal status class σ is the form of uptake requested for a claim: irrelevant, authenticated, admissible-only, evidence-compatible, probative-with-limits, method-valid within domain, expert-opinion-admissible, likelihood-supporting, source-consistent, source-identifying, causally supportive, legally actionable, element-sufficient under a declared procedural posture, factfinder-preserved, blocked, or failed.*

The taxonomy is not offered as a substitute for jurisdictional categories. It is a cross-jurisdictional audit vocabulary. A particular court or system may collapse, refine, or rename statuses. AASC requires only that the status being asserted be declared.

Definition 3.4 (Outcome relation). *An outcome relation is not an evidentiary status. It records whether a claim is non-outcome-bearing, element-supporting, element-sufficient under a declared burden and procedural posture, remedy-supporting, or procedurally outcome-determinative. Outcome*

relation is a separate registry coordinate because a claim's evidentiary value cannot be converted into case disposition without legal elements, burden, posture, and procedural rules.

3.3 Status-vector model and local promotion

An earlier version of the architecture used a single default status ladder. That presentation is too linear for law. Legal statuses are not naturally ordered as one scalar sequence. An item can be authenticated yet hearsay-limited; admissible yet weak; method-valid in general yet unreliable as applied; epistemically probative yet excluded for prejudice, privilege, confrontation, or policy; or preserved for the factfinder rather than classified as failed. The present version therefore uses a status-vector model.

Definition 3.5 (Legal status vector). *The status of a claim under legal audit basis $\mathcal{A}_J$ is recorded as*

$$\text{Reg}_J(c) = (\text{TargetStatus}, \text{CarrierStatus}, \text{AdmissibilityStatus}, \text{ProbativeStatus}, \\ \text{MethodStatus}, \text{NetworkStatus}, \text{BurdenStatus}, \text{OutcomeRelation}, \text{ReportStatus}, \\ \text{Blockers}, \text{NextAction}).$$

Each coordinate records a legally distinct dimension of the claim. No coordinate may be silently converted into another.

Table 4: Status-vector coordinates.

Coordinate	Audit role
TargetStatus	What legal or evidentiary object is asserted: relevance, authentication, source identity, causation, admissibility, liability, or another target.
CarrierStatus	Whether the offered support is the right kind of carrier for the requested target and status.
AdmissibilityStatus	Whether the item is admissible, inadmissible, admissible for limited purpose, excluded for non-epistemic reasons, or jurisdictionally unresolved.
ProbativeStatus	What inferential force is licensed: compatible-only, probative-with-limits, likelihood-supporting, source-consistent, source-identifying, causal-supportive, or factfinder-preserved.
MethodStatus	Whether the method is unvalidated, valid only in a declared domain, valid but misapplied, reliably applied, or not method-dependent.
NetworkStatus	Whether supports are independent, contaminated, common-source, merely compatible, or globally corroborative.
BurdenStatus	Which burden or threshold the carrier can reach, if any; lower-burden satisfaction cannot be reported as higher-burden satisfaction.

Coordinate	Audit role
OutcomeRelation	Whether the claim is non-outcome-bearing, element-supporting, element-sufficient under posture P , remedy-supporting, or procedurally outcome-determinative.
ReportStatus	The strongest permissible wording without target, carrier, burden, scope, uncertainty, or factfinder-role drift.
Blockers	Active reasons stronger statuses are unavailable.
NextAction	The minimal operation that would discharge at least one load-bearing blocker.

Definition 3.6 (Local status promotion). *A status promotion is local when it moves along one declared coordinate or between two explicitly related coordinates: source-consistency to source-identification, admissible-only to probative-with-limits, method-valid-in-general to reliably-applied-in-case, compatible narrative to causation, or computation-only to validated machine-evidence status. A promotion is licensed only when the blocker separating the lower status from the higher status is discharged.*

3.4 Local promotion edges

The protocol does not require a global ranking of all legal statuses. It audits local transitions. The following are canonical promotion edges:

authenticated $\nrightarrow$ substantively true, admissible $\nrightarrow$ proven,

qualified expert $\nrightarrow$ reliable method, method valid generally $\nrightarrow$ reliably applied here,

similarity $\nrightarrow$ source identity, likelihood relation $\nrightarrow$ categorical attribution,

machine output $\nrightarrow$ warrant for the asserted legal target, narrative coherence $\nrightarrow$ legal causation.

A laundering route exists when a report crosses one of these edges without discharging the blocker that separates the lower and higher statuses. The local-promotion model is the core legal upgrade over a scalar status ladder: it allows authentication, admissibility, method validity, probative use, burden satisfaction, factfinder preservation, and outcome relation to be audited separately.

Definition 3.7 (Local-promotion graph). *For a jurisdictional ledger J , let*

$$G_J = (V_J, E_J, B_J),$$

where V_J is the set of legal or evidentiary status nodes, E_J is the set of declared local promotion edges, and $B_J(e)$ assigns to each edge $e : s_i \rightarrow s_j$ the blocker that must be discharged before the report may move from s_i to s_j . A laundering route occurs when a report asserts movement along an edge $s_i \rightarrow s_j$ without discharging $B_J(e)$.

Table 5: Illustrative local-promotion edges.

Local edge	Required discharge	Typical failure
Authentication $\rightarrow$ substantive truth	Truth-purpose carrier, hearsay/proxy disposition, completeness, and context.	Authentication laundering.
Admissibility $\rightarrow$ proof	Burden, factfinder role, probative force, and alternative disposition.	Admissibility laundering.
Expert qualification $\rightarrow$ reliable method	Method reliability and fit to target.	Authority laundering.
General method validity $\rightarrow$ reliable case application	Case-specific application, data sufficiency, and limits.	Application laundering.
Similarity $\rightarrow$ source identity	Comparison class, method validation, error, alternatives, and report limits.	Source-identity laundering.
Likelihood relation $\rightarrow$ categorical attribution	Legal standard and report carrier for categorical assertion.	Probability laundering.
Machine output $\rightarrow$ warrant for the asserted legal target	System identity, data provenance, validation, error, domain, and report language.	Computation laundering.
Narrative coherence $\rightarrow$ causation	Causal carrier, alternative causes, legal standard, and damages relation.	Narrative laundering.

The graph’s role is not to rank legal statuses globally. It identifies the precise transition being claimed and the blocker that must be discharged before the report may move across that edge.

The vector model preserves the legal role of factfinders. Factfinder-preserved status applies when the audit verifies that threshold presentation conditions have been met, while doctrine assigns weight, credibility, or ultimate inference to the jury, judge, tribunal, or administrative factfinder. In that posture the validator does not return a disguised verdict. It returns a bounded report: the claim may be presented only in language that preserves the factfinder’s legally assigned role. If doctrine assigns weight, credibility, or ultimate inference to the factfinder after a threshold showing, the correct audit status may be *factfinder-preserved*. That status is not a failure. It records that the protocol may police report language and carrier adequacy but may not replace the legally assigned decision-maker.

Table 6: Factfinder-preserved boundaries.

Legal function	AASC treatment
Admissibility threshold	Jurisdictional gate; the audit records whether the carrier satisfies the declared presentation threshold.
Weight	Factfinder-preserved unless the legal basis assigns weight to a gatekeeper or specialized tribunal.
Credibility	Factfinder-preserved except where doctrine creates a threshold credibility or reliability gate.

Legal function	AASC treatment
Sufficiency	Burden/status coordinate; the audit records whether the carrier can reach the declared threshold without deciding facts beyond its basis.
Ultimate finding	Outside the AASC certificate unless a declared legal standard and procedural posture make the target element-sufficient or outcome-relevant.

Protocol Result 3.1 (Unique registry assignment under a status vector). *For a completed legal registry row with a declared status-vector structure, the audit returns a determinate vector, blocker set, and next-action field. It need not return a single globally ranked status.*

Verification. The registry row fixes the target, carrier, jurisdiction, admissibility posture, method status, evidence network, burden standard, report object, and blockers. Each coordinate is evaluated under its own declared standard. If a coordinate is unresolved, the blocker is recorded rather than collapsed into a scalar failure. Local promotions are allowed only where a corresponding blocker has been discharged. Therefore the audit has a determinate output without requiring a total order over heterogeneous legal statuses. $\square$

4 Doctrinal crosswalk and legal sovereignty

The legal-domain version must be doctrine-facing. AASC is not a free-floating replacement for evidence law. It is a warrant protocol that records how doctrine-supplied standards enter the target, carrier, alternative-disposition, laundering, and report coordinates. The crosswalk in Table 7 is therefore central rather than ornamental.

Table 7: Doctrinal crosswalk for AASC legal audit.

Doctrine or practice	Legal role	AASC coordinate
Relevance	Connects an item to a legal proposition or element.	Target alignment; the evidence must be about the asserted legal target.
Rule 901-type authentication	Establishes that the item is what the proponent claims it is.	Item-identity target fixation; not substantive truth.
Rule 702-type expert evidence	Controls qualification, helpfulness, sufficient facts/data, reliable principles/methods, and reliable application.	Expert carrier adequacy and report preservation.
Daubert/Kumho reliability	Gatekeeping for scientific, technical, or specialized knowledge.	Method-carrier audit and domain-of-use control.

Doctrine or practice	Legal role	AASC coordinate
Rule 403-type balancing	Limits or excludes use despite relevance.	Report/use coordinate; prejudice and confusion can block otherwise probative use.
Hearsay	Restricts use of out-of-court statements for truth purposes.	Carrier/proxy problem; semantic content is not enough if the legally required carrier is missing.
Chain of custody	Preserves evidence identity and integrity through transfer.	Provenance carrier; chain support is not source identity or causation.
Burden of proof	Determines the threshold for requested legal uptake.	Requested status class and burden coordinate; lower-burden warrant cannot be reported as higher-burden satisfaction.
Factfinder role	Allocates credibility, weight, and ultimate inference.	Factfinder-preserved status; the audit preserves boundaries rather than deciding weight.
Expert independence/impartiality	Controls whether expert assistance is fair, objective, and non-partisan.	Carrier-integrity and evidence-network coordinates; blocks authority laundering.
Forensic validation	Establishes method limits, error profile, standards, and domain of use.	Method-status and report-language coordinates.
Machine-generated evidence	Tests whether system output can carry legal status without ordinary expert testimony.	Computation-laundering audit: system identity, input provenance, validation, error profile, and report limit.

The crosswalk also marks where AASC must stop. If the jurisdiction excludes a claim for privilege, confrontation, constitutional violation, unfair prejudice, discovery sanction, or public policy, the admissibility coordinate is controlled by law even if the epistemic carrier is strong. Conversely, if a permissive rule admits a claim, the audit may still report it as weak, compatible-only, factfinder-preserved, or overreported if the proponent’s language exceeds the carrier.

5 Local promotion graph and report boundaries

The paper does not use a global legal status ladder. Legal statuses form a vector, and the central audit object is a local promotion edge. A laundering route exists when a report moves from one status to another without discharging the blocker separating them. The following edges are typical:

authenticated $\not\Rightarrow$ *substantively true*,

admissible $\not\Rightarrow$ *proof of the asserted target*,

qualified expert $\nRightarrow$ *reliable method*,
reliable method generally $\nRightarrow$ *reliable application here*,
similarity $\nRightarrow$ *source identity*,
likelihood relation $\nRightarrow$ *categorical source attribution*,
machine output $\nRightarrow$ *evidentiary proof*,
narrative coherence $\nRightarrow$ *legal causation*,
lower burden satisfaction $\nRightarrow$ *higher burden satisfaction*.

Source status		Promoted report	Required discharge
Authenticated item	$\nRightarrow$	Substantive truth	truth-bearing carrier
Admissible evidence	$\nRightarrow$	Proof of asserted target	burden/status carrier
Qualified expert	$\nRightarrow$	Reliable method	method validation
Method valid generally	$\nRightarrow$	Reliable application here	case-specific application
Similarity	$\nRightarrow$	Source identity	source-attribution carrier
Machine output	$\nRightarrow$	Warrant for asserted target	system-validation carrier
Narrative coherence	$\nRightarrow$	Legal causation	causal carrier

Figure 1: Local promotion graph, shown as blocked illustrative edges. Each blocked edge may be crossed only when the registry records the carrier discharge associated with that edge.

Definition 5.1 (Local promotion graph). *A local promotion graph for a legal audit basis is a labeled directed graph*

$$G_J = (V_J, E_J, B_J),$$

where V_J is the set of legal or evidentiary status nodes, E_J is the set of proposed local promotion edges, and B_J assigns to each edge the blocker or carrier condition that must be discharged before the promoted report may be made.

Definition 5.2 (Local promotion edge). *A local promotion edge is a proposed transition from a lower, narrower, or different legal status to a stronger or differently targeted status. The edge is licensed only if the carrier discharges the blocker that distinguishes the two statuses under the declared legal audit basis.*

Protocol Result 5.1 (No unregistered local promotion). *A legal report may not cross a local promotion edge unless the registry records the target, carrier, alternative disposition, laundering discharge, burden threshold, and report wording that license the crossing.*

Verification. A local promotion changes the legal status being asserted. If the registry does not record the condition that separates the source status from the target status, then the promoted

report treats an unresolved blocker as if it were discharged. That is precisely legal laundering. Therefore the report must either remain at the lower status, preserve the blocker, or supply the missing carrier field. $\square$

5.1 Epistemic claim-status warrant is not legal admissibility

AASC is an epistemic and report-preservation protocol for claim status. Legal admissibility is a jurisdictional act governed by doctrine, procedure, judicial discretion, constitutional constraints, and institutional role. These are related but not identical.

An item may be epistemically strong yet legally inadmissible because of privilege, confrontation, hearsay, unfair prejudice, constitutional violation, discovery sanction, policy exclusion, or procedural defect. Conversely, an item may be legally admissible while remaining epistemically weak, compatible-only, or report-limited. AASC does not erase either possibility. It records the distinction in the status vector.

Doctrinal Interface Proposition 5.1 (Epistemic warrant does not entail admissibility). *A claim may satisfy AASC target, carrier, and report-preservation conditions for an epistemic status without being legally admissible under J; and a claim may be legally admissible under J without satisfying stronger AASC carrier conditions for proof, source identity, or causation.*

Verification. Admissibility is a legal status determined by a jurisdictional rule set. Epistemic claim-status warrant concerns target fixation, carrier adequacy, alternative disposition, laundering, and report language. The two classifications have overlapping but non-identical coordinates. If a non-epistemic exclusionary rule applies, the admissibility coordinate can be negative despite epistemic carrier strength. If a permissive admissibility threshold is met, the admissibility coordinate can be positive while probative, source-identifying, causal, element-sufficiency, or outcome-relation coordinates remain limited. Therefore neither classification entails the other. $\square$

5.2 Legal audit objects

Definition 5.3 (Legal claim). *A legal claim c is an assertion offered for legal, evidentiary, procedural, forensic, causal, institutional, adjudicative, or legally actionable uptake. The claim may be factual, legal, mixed, expert, forensic, documentary, testimonial, computational, statistical, or procedural.*

Examples include: “Trace X came from suspect Y ”; “Expert E used a reliable method”; “Document D authenticates transaction T ”; “Witnesses A and B independently corroborate event P ”; “Machine output O establishes legal fact F ”; “Act A caused harm H ”; “Evidence item E is admissible for purpose P ”; and “The defendant is liable under standard S .”

Definition 5.4 (Legal target). *The legal target of c is the object, relation, status, event, inference, element, or legal consequence that c purports to assert. Legal targets include relevance, authenticity, chain preservation, admissibility, probative force, credibility, method reliability, expert assistance, likelihood under a population, source consistency, source identity, causation, intent, legal characterization, liability, guilt, remedy, actionability, and outcome relation.*

Definition 5.5 (Legal carrier). *The legal carrier of c is the structure that purports to give c warrant at the requested status. It may be testimony, a document, a chain-of-custody ledger, physical*

evidence, metadata, a forensic method, an expert report, a statistical model, a likelihood ratio, an authentication record, a statutory text, a precedent, a machine-output ledger, or a procedural record.

Criterion 5.1 (Legal carrier adequacy test). *A carrier is adequate for legal status σ only if it satisfies the following eight tests relative to the declared audit basis:*

1. **Target alignment:** *it supports the same target asserted by the claim.*
2. **Status-role adequacy:** *it is the right kind of carrier for the requested status.*
3. **Jurisdictional adequacy:** *it satisfies the declared legal or procedural standard.*
4. **Method adequacy:** *where method is material, the method is validated, accepted, or otherwise licensed under the declared standard.*
5. **Application adequacy:** *where method is applied to facts, the application is adequate for the status asserted.*
6. **Provenance adequacy:** *evidence identity, chain, source, and preservation are fixed or their limits are reported.*
7. **Alternative adequacy:** *admissible rival hypotheses are disposed of as excluded, immaterial, defeated under the burden standard, factfinder-preserved, declared unresolved, or preserved in the report.*
8. **Report adequacy:** *the final legal language does not exceed the status licensed by the carrier.*

Failure of a test yields target-unfixed, carrier-unfixed, authenticated-only, admissible-only, evidence-compatible, probative-with-limits, domain-dependent, laundered, or failed status according to the load-bearing blocker.

Definition 5.6 (Legal report object). *The legal report object $R_J(c)$ is the strongest statement that may be responsibly issued about c without changing its audited status. It records target, carrier, jurisdiction, standard, scope, limitations, uncertainty, burden, permissible use, blockers, and nonclaim boundaries.*

The report object is not decorative. Legal consequences often turn on wording. “Consistent with,” “cannot exclude,” “more likely under hypothesis H_1 than H_2 ,” “admissible for limited purpose P ,” “probative subject to limitation L ,” “method-valid within domain D ,” “source-identifying under standard S ,” and “element-sufficient under procedural posture P ” are different legal statuses.

5.3 Legal target fixation

Protocol Result 5.2 (Target fixation as a necessary audit condition). *A legal claim cannot receive a requested legal status σ unless its legal target is fixed up to legally harmless variation.*

Verification. A legal status is a status of an asserted target under a jurisdictional and procedural ledger. If the target is unfixed, the same words can refer to different legal objects: source consistency rather than source identity, admissibility rather than proof, qualification rather than method reliability, authentication rather than substantive truth, temporal sequence rather than legal causation, or lower-burden satisfaction rather than higher-burden satisfaction. A status assignment made before target fixation would attach to a moving object. Therefore target fixation is necessary for any stable legal status assignment. $\square$

5.4 Legal target taxonomy

The following target families recur across legal audits.

Table 8: Legal targets and common drift.

Target family	Common drift
Existence	An event or item existed; drift occurs when existence is reported as source or causation.
Authentication	Item is what proponent claims; drift occurs when authenticity is reported as truth of contents.
Chain preservation	Item continuity is preserved; drift occurs when preservation is reported as source identity.
Relevance	Evidence has a tendency to affect a fact of consequence; drift occurs when relevance is reported as sufficiency.
Admissibility	Evidence may be received for a purpose; drift occurs when admissibility is reported as proof.
Probative force	Evidence supports an inference; drift occurs when limited probative status is reported as closure.
Method reliability	A method is fit for a domain; drift occurs when general validity is reported as case-specific correctness.
Expert assistance	Opinion may assist factfinder; drift occurs when assistance is reported as legal conclusion.
Source consistency	Trace is compatible with source; drift occurs when compatibility is reported as identification.
Source identity	Source is asserted; drift occurs when likelihood language is treated as categorical certainty.
Causation	Conduct caused harm under legal standard; drift occurs when chronology or narrative fit is reported as causation.
Legal entitlement	A rule licenses a remedy or defense; drift occurs when plausible argument is reported as established entitlement.

5.5 Examples of target drift

A DNA or pattern-comparison statement can shift from match, inclusion, likelihood, or cannot-exclude language into categorical source identification. A toolmark or handwriting comparison can shift from observed similarity into uniqueness. A witness-convergence claim can shift from agreement to independent corroboration. An expert report can shift from qualified opinion to reliable application. An AI output can shift from generated classification to validated proof of the asserted legal target. A document can shift from authenticity to truth of contents. A statute can shift from legal authority to factual proof. AASC treats these not as stylistic differences but as target changes.

6 Legal carriers, freedoms, and laundering

Protocol Result 6.1 (Carrier adequacy is status-relative). *A carrier adequate for one legal status is not automatically adequate for any stronger status.*

Verification. A carrier bears only the legal status for which it is adequate under the declared audit basis. Chain of custody may support provenance but not source identity. Expert credentials may support qualification but not method reliability. A validated method may support admissibility but not reliable application to case-specific facts. A likelihood ratio may support a likelihood relation but not categorical source assertion. Authentication may support item identity but not substantive truth. If the same carrier is used to report a stronger target without discharging the additional status condition, a status transition has occurred without carrier support. That is carrier laundering. Therefore carrier adequacy is status-relative. $\square$

6.1 Carrier-status matrix

Table 9: Legal carriers and status boundaries.

Carrier	May support	Does not alone support
Chain-of-custody record	preservation, provenance, continuity	source identity, causation, truth of contents
Expert credentials	qualification, experience basis	method validity, reliable application, correctness
Validated method	method reliability within domain	case-specific conclusion without application audit
Statistical likelihood ratio	comparative likelihood under assumptions	categorical source certainty
Witness testimony	testimonial assertion, direct observation if credited	independent convergence without network audit
Document metadata	authenticity, timing, custody clues	substantive truth or legal effect by itself
AI or machine output	computation, classification, generated record	reliability, source identity, or proof without validation ledger
Statute or precedent	legal rule source, authority	factual truth of disputed evidence
Lab accreditation	institutional quality marker	correctness of a specific test or opinion

6.2 Carrier failure modes

Common carrier failures include wrong target, incomplete chain, invalid method, valid method outside domain, unreported error rate, undeclared comparison class, unsupported expert inference, inadmissible proxy carrier, black-box machine output, missing model identity, improper application, and report language exceeding the carrier. UEAP’s contribution is not that these are unknown to

lawyers. Its contribution is that they are recorded as blockers that prevent a stronger status from being silently asserted.

6.3 Legal freedom ledger

Definition 6.1 (Legal freedom ledger). *The legal freedom ledger of claim c is*

$$\mu_J(c) = \{\mu_{\text{target}}, \mu_{\text{carrier}}, \mu_{\text{jurisdiction}}, \mu_{\text{procedure}}, \mu_{\text{chain}}, \mu_{\text{method}}, \\ \mu_{\text{population}}, \mu_{\text{comparison}}, \mu_{\text{witness}}, \mu_{\text{causal}}, \mu_{\text{machine}}, \mu_{\text{report}}\}.$$

A component is active when admissible variation of that type can change the claim’s legal target, carrier, status, scope, or report object.

Table 10: Legal freedom components.

Freedom	Audit question
Target	Can the asserted legal object shift?
Carrier	Can the warrant-bearing evidence or method change?
Jurisdiction	Does a different legal forum or rule change the status?
Procedure	Does procedural posture change the permissible use?
Chain	Can item identity, custody, preservation, or alteration change?
Method	Does validation, domain, error model, or application remain open?
Population	Does the base rate or comparison class change the inference?
Comparison	Are alternative sources or hypotheses still admissible?
Witness	Are witness sources independent, contaminated, coached, or narrative-linked?
Causal	Can alternative causal paths, intervening causes, or baseline conditions change the inference?
Machine	Can model identity, data provenance, version, threshold, or validation alter the output status?
Report	Does legal wording, burden, or permissible use change the claim?

Protocol Result 6.2 (Legal freedom disposition criterion). *A legal claim is fixed at requested status σ only when every status-changing legal freedom is disposed of as skin, legally inadmissible, legally immaterial, declared input, obstruction-only, discharged by the carrier, factfinder-preserved, or preserved as a blocker. If any content-changing legal freedom remains hidden, the claim cannot receive claim-status warrant at σ .*

Verification. If a legal freedom can change the claim’s target, carrier, scope, evidentiary status, or report object, then more than one non-skin legal realization remains available. A report at a single stronger status treats the freedom as resolved. The freedom is harmless only if it is skin; excluded

from the legal domain; declared as a conditioning input; negative only; discharged by the carrier; or preserved in the blocker set and report. These possibilities exhaust the legally responsible ways to handle an active freedom. Therefore hidden active freedom blocks final warrant at the requested status. $\square$

6.4 Legal laundering taxonomy

Definition 6.2 (Legal laundering). *A legal claim is laundered when a transformation changes the apparent legal status of the claim while leaving unresolved the blocker that separates the original status from the reported status.*

The transformation may be rhetorical, expert, institutional, documentary, computational, statistical, procedural, or narrative. It need not be intentional. A report can launder a claim by compressing important status boundaries into ordinary language.

Table 11: Legal laundering routes.

Route	Legal manifestation	Required discharge
Authority laundering	Expert, court, lab, agency, or institutional status replaces the carrier.	Identify target-aligned carrier.
Method laundering	Procedure label replaces validation or reliable application.	Establish method and application adequacy.
Chain laundering	Assumed preservation replaces provenance.	Fix custody, identity, and alteration ledger.
Similarity laundering	Similarity is reported as source identity.	Fix comparison class, error model, alternatives, and report standard.
Narrative laundering	Coherent story is treated as proof or causation.	Close alternatives under legal standard.
Witness-network laundering	Contaminated or common-source witnesses are counted as independent.	Audit independence and contamination.
Baseline laundering	Population, base rate, or comparison class is hidden.	Declare and justify population/comparison class.
Burden laundering	Lower-burden result is worded as higher-burden satisfaction.	Discharge higher-burden status.
Admissibility laundering	Admitted evidence is reported as proof.	Preserve admissibility/proof distinction.
AI/computation laundering	Machine output is treated as reliable legal evidence.	Fix system, data, validation, error, domain, and report.
Authentication laundering	Authentic item is treated as substantively true.	Provide substantive truth carrier.

Route	Legal manifestation	Required discharge
Causation laundering	Sequence, correlation, or fit is treated as legal causation.	Supply causal carrier and alternative disposition.
Precedent laundering	Legal authority is treated as factual support.	Separate legal rule target from factual target.

Protocol Result 6.3 (Legal laundering exclusion). *No transformation may increase a legal claim’s status unless it discharges the blocker separating the lower status from the requested higher status. Otherwise the transformed claim inherits the original blocking status.*

Verification. A status increase is licensed only when a condition required for the stronger status has been satisfied. If a transformation changes wording, presentation, expert packaging, institutional source, statistical form, or computational format without satisfying that condition, the unresolved alternative or carrier defect remains. Reporting the transformed claim at the higher status treats a remaining blocker as discharged. That is laundering. Therefore the transformed claim inherits the original blocking status unless the transformation explicitly discharges the blocker. $\square$

7 Evidence networks and factfinder-preserved status

Evidence is not additive by appearance. Multiple sources can converge because they independently constrain a target, or because they share contamination, prompting, media exposure, investigative narrative, common data, common laboratory process, shared model assumptions, common expert interpretation, or a common base-rate error. A legal audit therefore treats evidence as a network.

Definition 7.1 (Legal evidence network). *The legal evidence network $Net_J(c)$ is a graph whose nodes are witnesses, documents, physical items, machine outputs, expert reports, metadata, admissions, circumstantial facts, statistical models, and procedural records; and whose edges record dependence, communication, contamination, common source, common method, common database, shared narrative, shared expert, or shared report transformation.*

Definition 7.2 (Independent legal support). *Evidence nodes are independent relative to c only if they do not share a dependence that matters to the target: witness contamination, common source, coaching, media exposure, shared database, shared model, common lab process, same expert interpretation, hidden proxy, or reportability-transforming baseline.*

Protocol Result 7.1 (Legal evidence non-amplification). *Multiple evidence supports do not promote a claim across a legal status boundary merely by accumulation. Promotion is licensed only when the evidence network is target-aligned, sufficiently independent under the declared standard, globally compatible, and jointly closes the relevant alternatives or satisfies the jurisdictional evidentiary standard for σ .*

Verification. Accumulation can increase legal weight only to the extent that the additional supports add independent constraint relevant to the target. If sources share contamination, proxy structure, method dependence, or narrative ancestry, adding them does not close the relevant alternatives. If sources are only pairwise compatible, a global alternative may remain. If the jurisdictional standard

requires closure of a particular blocker and the network leaves that blocker active, status promotion is unlicensed. Therefore evidence accumulation can promote status only through target-aligned and legally adequate network closure. $\square$

7.1 Witness convergence audit

Witness convergence is a useful stress test because the surface signal of agreement can hide very different evidentiary structures. The audit asks:

1. Did witnesses observe independently?
2. Did witnesses communicate before reporting?
3. Were they exposed to common media, police theory, counsel framing, or community narrative?
4. Were accounts elicited through similar prompts or leading questions?
5. Are similarities generic, expected, or discriminating?
6. Are inconsistencies material, harmless, or target-changing?
7. What target is actually supported: occurrence, sequence, identity, intent, causation, or credibility?

A witness-network claim can receive independent-corroboration status only if independence itself is a warrant-bearing carrier. Otherwise the status is compatible-only, contaminated convergence, narrative convergence, evidence-compatible, or failed.

8 Forensic source-attribution audit

8.1 Forensic source-attribution claim

The canonical forensic claim is:

Trace X identifies source Y .

This claim is stronger than compatibility, similarity, cannot-exclude language, or a likelihood relation. Source-identifying status requires a carrier that fixes item identity, method validity, case-specific application, comparison class, error-rate discipline, alternative-source treatment, and report language.

8.2 Required carrier

A source-identification carrier includes at least the following fields: item identity, chain of custody, validated method, comparison class, population or relevant source universe where applicable, error model, laboratory protocol, analyst procedure, verification practice, alternative-source ledger, analyst-discretion ledger, and report-language constraint.

A publication-ready forensic audit must also be method-specific. For cartridge-case and other toolmark comparisons, the legal status turns on technical features that cannot be collapsed into generic similarity.

Table 12: Method-specific cartridge-case issues and AASC roles.

Cartridge-case issue	AASC role
Class characteristics	Target narrowing: class agreement may support compatibility but not individual source identity.
Individual marks	Candidate carrier for stronger source-consistency or source-attribution status, subject to validation and application limits.
Subclass characteristics	Alternative-source blocker: marks shared by manufacturing process can mimic individualizing agreement.
Consecutively manufactured barrels or tool surfaces	Comparison-class and alternative-source pressure on uniqueness or discrimination claims.
Analyst discretion	Carrier/application freedom; requires documentation, standards, and report limits.
Contextual information	Evidence-network contamination risk; may affect examiner interpretation.
Proficiency testing	Method/application support; does not alone prove correctness in the case.
Error-rate or black-box studies	Carrier adequacy and report-language limit; error evidence constrains categorical claims.
Blind or independent verification	Network-independence support; not automatic global closure.
Inconclusive category handling	Report-preservation requirement; exclusion of inconclusives can distort performance claims.
Categorical identification language	Local-promotion risk: source-consistency or support-for-common-source language may be laundered into categorical attribution.

The audit does not decide the validity of firearms or toolmark identification as a field. It specifies what must be supplied before a report may move from similarity, source-consistency, or support-for-common-source language to source-identifying language.

Legal Audit Lemma 8.1 (Similarity does not license source-identity status). *A similarity relation between trace X and source Y does not license a source-identification claim unless the carrier disposes of comparison-class, population, method-validity, error-rate, chain, analyst-discretion, and alternative-source freedoms under the requested legal status.*

Verification. Similarity fixes only a relation between observed features. Source identity asserts an origin relation. The move from feature similarity to origin requires additional conditions: the item must be the relevant item, the method must be reliable for the comparison class, the application must be reliable in the case, error rates and limitations must be preserved, and relevant alternative sources must be disposed of under the legal standard. If those conditions remain open, a similarity report may be compatible, source-consistent, or likelihood-supporting, but not source-identifying.

Reporting it as source identity would change the target and status without discharging the blockers.
 □

8.3 Detailed worked audit: cartridge-case comparison

The detailed audit uses a hypothetical cartridge-case comparison because it exposes most forensic status-promotion risks without depending on a real case.

Hypothetical fact pattern. A cartridge case C is recovered from a shooting scene. Police later recover firearm F from suspect S . A firearms examiner compares microscopic marks on C to test-fired cartridge cases from F and reports: “Cartridge case C was fired from firearm F .” The proponent offers this conclusion to support source identity at trial.

Step 1: literal claim. The literal claim is not merely that C and the test fires share marks. It is a source-identification claim: F is asserted to be the firearm that fired C .

Step 2: requested status. The requested status is source-identifying forensic evidence with probative use on a disputed fact. If the report is used only as “consistent with F ,” the requested status is lower. If it is used to prove that S committed the shooting, a further actor-attribution and causation chain is required.

Step 3: jurisdictional ledger. The audit must declare the jurisdiction, evidence rules, expert-testimony standard, admissibility posture, burden context, permissible use, and factfinder role. In a U.S. federal-style ledger, Rule 702-type reliability and reliable-application requirements govern the expert carrier, Rule 403-type concerns may limit use, and authentication/chain rules govern the physical item. In a Canadian-style ledger, Mohan and White Burgess type criteria enter the expert-carrier and independence fields.

Step 4: target fixation. The target must be one of the following, not all at once:

1. feature similarity between C and test fires from F ;
2. source consistency: F cannot be excluded or is compatible with C ;
3. likelihood-supporting relation under a declared source population and method;
4. source identification: C was fired from F ;
5. actor attribution: suspect S fired the shot;
6. legal guilt or liability.

Only the fourth is the source-identification target. The fifth and sixth require additional carriers.

Step 5: carrier inventory. The carrier includes: recovery and packaging record for C ; storage and transfer chain; recovery and identity record for F ; test-fire protocol; microscope/comparison procedure; examiner qualifications; method-validation literature; laboratory quality controls; verification practice; known limitations; error-rate evidence; subclass-characteristic controls; analyst notes; report wording; and disclosure of uncertainty.

Step 6: active freedoms. The legal freedom ledger records at least: chain freedom, item-identity freedom, method-validation freedom, application freedom, examiner-discretion freedom, comparison-class freedom, subclass-characteristic freedom, error-rate freedom, alternative-source freedom, and report-language freedom.

Step 7: alternative disposition. Rival hypotheses need not be metaphysically eliminated. They must be disposed of relative to the requested legal status. Alternatives include: C was fired by another firearm with similar marks; subclass characteristics explain similarity; the recovered C is not the relevant item; F identity or chain is compromised; the method is reliable only for a weaker status; examiner discretion or contextual bias affects the conclusion; or the conclusion is admissible but should be stated in limited language. Each alternative is excluded, rendered immaterial, defeated under the burden, factfinder-preserved, left open as report-limiting, or fatal to source-identifying status.

Step 8: laundering risks. The principal laundering routes are:

- similarity-to-source-identity laundering;
- examiner-confidence-to-method-validity laundering;
- laboratory-accreditation-to-case-correctness laundering;
- admissibility-to-proof laundering;
- chain-assumption-to-item-identity laundering;
- likelihood/support-to-categorical-certainty laundering;
- report-language laundering when “consistent with” is paraphrased as “matched” or “identified.”

Step 9: status-vector assignment. The registry may return different coordinate values. If chain is intact and the comparison is properly conducted but the method carrier supports only limited language, the ProbativeStatus may be source-consistent or likelihood-supporting rather than source-identifying. If the method is admitted for factfinder weighing, ReportStatus may be factfinder-preserved with strict wording. If chain or item identity fails, CarrierStatus may be failed or carrier-unfixed. If the report says “ F fired C to the exclusion of all other firearms” without carrier support for that certainty, ReportStatus is overreported and the source-identifying claim is blocked.

Table 13: Detailed registry row for cartridge-case comparison.

Field	Audit entry
Claim ID	LEG-FOR-CART-001
Literal claim	Cartridge case C was fired from firearm F .
Requested status	Source-identifying forensic conclusion with probative trial use.
Jurisdiction ledger	Must declare evidence rule, expert standard, admissibility posture, burden context, factfinder role, and permissible use.
Target	Source identity of cartridge case C relative to firearm F ; not actor identity or guilt.

Field	Audit entry
Carrier	Chain for C and F ; test-fire protocol; examiner method; validation literature; application notes; error-rate/limitation report; verification; comparison class.
Active freedoms	Chain, item identity, method validation, application, comparison class, alternative firearms, subclass characteristics, analyst discretion, report wording.
Alternative disposition	Alternatives excluded, immaterial, defeated under burden, factfinder-preserved, report-limiting, or fatal.
Laundering risks	Similarity to identity; accreditation to correctness; admissibility to proof; confidence to method validity; likelihood to certainty.
Possible lower statuses	Authenticated-only; method-valid-within-domain; source-consistent; likelihood-supporting; probative-with-limits; factfinder-preserved.
Blocked overclaim	Categorical source identity if method, application, comparison class, and error profile do not support that report language.
Report-preserving form	“The observed marks are consistent with having been produced by firearm F under method M , with limitations L ,” unless a stronger source-identification carrier is affirmatively declared.
Next action	Declare jurisdictional standard, method validation basis, error/limitation profile, comparison class, chain status, alternative-source disposition, and exact report language.

Audit result. The cartridge-case claim does not automatically fail. Nor does it automatically become source identification because an examiner used that label. The AASC result depends on the declared carrier. With intact chain, sufficient method validation, reliable case application, disciplined alternative disposition, and report-preserving wording, the claim may receive a strong forensic status under J . Without those fields, the strongest responsible status is lower: source-consistent, likelihood-supporting, probative-with-limits, factfinder-preserved, or failed. The audit protects the proper evidentiary value of the comparison by preventing over-reporting.

8.4 Forensic report-language variants

The same technical comparison can be reported at several different legal statuses. The audit must therefore classify report language, not merely the underlying examination.

Table 14: Report-language variants for cartridge-case comparison.

Report language	Likely warrant status	Required caution
“The observed marks are consistent with firearm F .”	Source-consistent or compatible-only.	Does not identify F as source.

Report language	Likely warrant status	Required caution
“The evidence supports firearm F as the source.”	Support-for-source claim.	Requires declared support scale, comparison class, and limits.
“The evidence strongly supports firearm F as the source.”	Strong support under declared method and scale.	Requires validation, error/limitation report, and no categorical paraphrase.
“Cartridge C was fired from firearm F .”	Source-identifying claim.	Requires source-identification carrier under J .
“To the exclusion of all other firearms.”	Categorical exclusion claim.	Usually requires a much stronger uniqueness/exclusion carrier; otherwise report-laundered.

8.5 Forensic-reporting guidance interface

The audit is compatible with forensic-reporting guidance that distinguishes categorical identification, source-attribution language, likelihood or support language, inconclusive results, error-rate disclosure, and limitation statements. Its purpose is not to decide the scientific validity of firearms/toolmark identification. It controls which report language is warranted by the declared carrier. If the carrier supports only similarity, the report must not use source-identifying language. If the carrier supports a likelihood or support conclusion under a declared comparison class, the report must preserve that scale and its limits. If the report asserts categorical exclusion of all other firearms, the carrier must supply a warrant for that stronger edge in the local-promotion graph.

9 Expert testimony, authentication, and causation

9.1 Expert testimony: target and carrier separation

The canonical expert claim is:

Expert E concludes C ; therefore C has claim-status warrant.

AASC disaggregates this into separate targets: E is qualified; E ’s testimony is relevant or helpful; E uses sufficient facts or data; E uses reliable principles and methods; E applies them reliably to the facts; C supports the asserted legal target; and the report language preserves uncertainty and scope.

In U.S. federal practice, Rule 702 as amended clarifies that the proponent must show, more likely than not, that the expert testimony satisfies the rule’s admissibility requirements and that each expert opinion stays within the bounds of what can be concluded from reliable application of the expert’s basis and methodology (Fed. R. Evid. 702, 2026; United States Courts, 2024). Daubert

supplies the gatekeeping frame for relevance and reliability, and Kumho extends that gatekeeping function to technical and other specialized knowledge (*Daubert*, 1993; *Kumho Tire*, 1999). In Canada, Mohan states relevance, necessity, absence of exclusionary rule, and proper qualification as admission criteria for expert evidence, while White Burgess emphasizes the expert’s duty to provide fair, objective, and non-partisan assistance (*Mohan*, 1994; *White Burgess*, 2015).

Table 15: Rule 702 decomposition as AASC carrier coordinates.

Rule 702 element	AASC coordinate
Qualified expert	Authority/role carrier; does not itself warrant the conclusion.
Helpfulness / fit	Target alignment and factfinder-use coordinate.
Sufficient facts or data	Carrier basis and evidentiary provenance.
Reliable principles and methods	Method-carrier adequacy.
Reliable application	Case-specific application carrier.
Opinion within reliable application	Report preservation; no method-to-overclaim promotion.
Rule 104(a) threshold	Burden/status coordinate for the admissibility gate.

Audit 9.1 (Expert overstatement). *Expert E is qualified and method M is generally reliable, but the final opinion C exceeds what M can support on the case-specific facts. The audit does not necessarily exclude all testimony. It returns a narrowed report: expert-opinion admissible or probative-with-limits only if C is rewritten to stay within the reliable application of the basis and methodology. If the overstatement remains, the report is laundered.*

Protocol Result 9.1 (Expert authority does not substitute for method warrant). *Expert qualification, institutional affiliation, laboratory status, publication record, or prior admissibility cannot substitute for the carrier required for the requested expert-testimony status unless the legal target is authority itself.*

Verification. Expert authority and method warrant are different targets. Authority may support qualification, experience, or institutional credibility. The requested status may require sufficient facts, reliable method, reliable application, independence, or fit. If authority is used to stand in for these carrier conditions, the claim is promoted without discharging method, application, or target-alignment blockers. Therefore expert authority cannot substitute for method warrant unless the claim under audit is itself a claim about authority. $\square$

Audit 9.2 (Expert testimony). *Claim: “Because expert E concludes C, C is established.” Target: legal or factual conclusion C. Carrier: expert report, facts/data, method, application, scope, and jurisdictional expert-testimony standard. Active freedoms: qualification, method validity, application, fit, independence, bias, uncertainty, and report wording. Laundering risks: credentials-to-correctness, admissibility-to-proof, general-method-to-case-application, confidence-to-reliability, and institutional-status-to-warrant. Final status may be expert-qualified-only, expert-opinion-admissible, method-valid within domain, probative-with-limits, evidence-compatible, or failed.*

9.2 Chain of custody and authentication targets

The canonical chain/authentication claim is:

Evidence item E is the same item collected at site S and is preserved for legal use.

Authentication and chain preservation are powerful but limited targets. Rule 901 in the U.S. requires evidence sufficient to support a finding that the item is what the proponent claims it is (Fed. R. Evid. 901, 2026). That target is not the same as the truth of every proposition later inferred from the item.

9.3 Authentication is not substantive truth

- physical identity;
- documentary authenticity;
- authorship;
- timestamp or metadata continuity;
- storage integrity;
- contamination exclusion;
- admissibility for a purpose;
- substantive truth of contents;
- source identity or causal relevance.

Protocol Result 9.2 (Authentication does not entail substantive truth). *Authentication or chain preservation may fix the identity or provenance of an evidence item, but it does not by itself establish the truth, causal significance, source identity, or legal effect asserted from that item.*

Verification. Authentication and chain preservation address whether an item is what it is claimed to be and whether its custody is sufficiently fixed for the relevant use. Substantive truth, source identity, causation, and legal effect are distinct targets requiring additional carriers. Treating authentication as proof of those targets changes the claim without adding the required carrier. That is authentication laundering. $\square$

Audit 9.3 (Document proof). *Claim: “Document D proves transaction T .” Target: transaction occurrence, not merely document identity. Carrier: document provenance plus substantive reliability carrier. Active freedoms: authorship, timestamp, alteration, hearsay or equivalent use rule, context, completeness, authority, and legal effect. Possible statuses: authenticated-only, admissible-only, probative-with-limits, evidence-compatible, or failed. Report boundary: an authenticated document may be reportable as authenticated; it is not reportable as proving the transaction without a substantive target-aligned carrier.*

9.4 Causation: sequence, mechanism, and legal cause

The canonical causation claim is:

Act A caused harm H .

Legal causation is not one target. Depending on jurisdiction and cause of action, the claim may involve temporal sequence, factual causation, but-for causation, material contribution, substantial factor, proximate cause, foreseeability, scope of liability, intervening cause, damages relation, or regulatory causation.

Protocol Result 9.3 (Narrative fit is not causal warrant). *A coherent reconstruction containing A before H does not license a causal legal claim unless the carrier closes alternative causal paths under the requested jurisdictional standard.*

Verification. Narrative coherence can establish that a sequence is possible or compatible with available facts. Causation asserts a stronger relation: that *A* caused *H* under a legal standard. Alternative causes, baseline conditions, intervening events, confounding, mechanism gaps, and damages allocation may remain open even when the narrative is coherent. If those alternatives are not closed or preserved as limitations, reporting narrative fit as causation promotes the claim without discharging causal blockers. Therefore narrative fit is not causal warrant. $\square$

Audit 9.4 (Legal causation). *Claim: “Defendant’s act caused plaintiff’s harm.” Target: jurisdiction-specific causation. Carrier: expert causal analysis, medical record, accident reconstruction, statistical evidence, eyewitness sequence, physical mechanism, and legal doctrine as applicable. Active freedoms: alternative cause, intervening cause, baseline condition, temporal ambiguity, mechanism gap, standard of proof, and damages relation. Final statuses: sequence-compatible, mechanism-compatible, causally-supportive, causally-established-under-standard, legally-actionable, or failed.*

10 AI and machine-generated legal evidence

10.1 Machine-generated evidence claim

The canonical machine-evidence claim is:

Machine-generated output *O* establishes legal fact *F*.

This is the clearest modern form of computation laundering. A machine output may be real, preserved, and useful. It may also be unauditable for the requested legal status if system identity, data provenance, validation, error profile, domain, and report language are not fixed.

Federal rulemaking materials show why this problem is live, while also showing why caution is required. Advisory Committee materials on proposed Rule 707 state that machine-generated evidence offered without expert testimony can raise reliability questions similar to Rule 702 expert issues, because Rule 702 is not obviously applicable where no expert witness is testifying (Advisory Committee on Evidence Rules, 2026). At the time of writing, the precise rulemaking status remains in motion; contemporary procedural-status reporting indicated that federal judicial rulemakers had delayed action for further expert consultation (Raymond, 2026). This paper treats machine evidence as an audit stress test, and proposed Rule 707 only as unsettled rulemaking material, not as enacted federal doctrine.¹

¹Proposed Rule 707 is cited only as rulemaking evidence that machine-generated evidence has become a recognized evidentiary reliability problem. The protocol does not depend on adoption of the proposed rule, and the paper makes no claim about its final content, timing, or enactment.

10.2 Machine-evidence typology

Machine evidence is not one category. Different machine-generated objects raise different carrier risks. The audit first identifies the kind of machine evidence before asking whether the carrier supports the requested legal target.

Table 16: AI and machine-evidence typology.

Evidence type	Primary AASC risk	Typical first status
Machine log or sensor record	Authentication, provenance, clock integrity, and chain of custody.	provenance-bound
Forensic software output	Method validation, error rate, input quality, and domain of use.	method-dependent
Risk score or predictive model	Calibration, proxy target, base rate, and actionability laundering.	calibrated or domain-dependent
Classifier output	Threshold choice, false positives/negatives, domain shift, and report language.	probabilistic support
Generative AI output	Source/provenance confusion, hallucination, prompt dependence, and unsupported factual assertion.	computation-only
Deepfake or synthetic media claim	Authenticity, fabrication, metadata, provenance, and forensic detection reliability.	authentication contested
AI-assisted expert report	Separation of expert carrier from machine carrier and reliable application.	hybrid expert-machine

10.3 Machine-evidence carrier

A machine-evidence carrier normally includes:

- system identity, version, configuration, and operator role;
- input data and data provenance;
- training exposure or benchmark exposure where relevant;
- validation regime and domain of validated use;
- error rates, uncertainty, calibration, and false-positive/false-negative costs;
- model threshold, sampling, prompt, tool access, or run condition;
- bias, robustness, and domain-shift analysis where material;
- human review and intervention history;
- chain of custody for inputs and outputs;
- report-language boundary.

Table 17: Machine-evidence targets and required carriers.

Machine-evidence target	Carrier needed
Output provenance	Logs, hashes, chain, system identity, version, and operator record.
Process reliability	Validation, error rates, domain testing, threshold policy, calibration, and robustness evidence.
Case-specific reliability	Input quality, operating conditions, configuration, operator actions, and domain fit.
Substantive fact assertion	Process reliability plus case-specific application plus target alignment and alternative disposition.
Legal reportability	Uncertainty, limitation, permissible-use language, and factfinder boundary.

NIST AI risk-management and bias-management materials are useful carrier-standard resources, especially where the legal target involves reliability, bias, robustness, or deployment context (National Institute of Standards and Technology, 2023; Schwartz et al., 2022). They do not themselves decide admissibility.

Protocol Result 10.1 (Machine output does not supply evidentiary warrant). *A machine-generated output does not confer claim-status warrant merely by being computed, recorded, or authenticated. It can support a requested legal status only when target, system identity, data provenance, validation, error profile, domain of use, and report object are fixed.*

Verification. A computation is an output of a system under inputs and conditions. Claim-status warrant at a requested status requires a target-aligned carrier. If system identity, data provenance, validation, error rates, domain, thresholds, or report boundaries remain open, the same output can support different statuses: computation-only, authenticated machine output, evidence-compatible, method-valid within domain, or probative-with-limits. Treating the output itself as proof collapses these statuses without discharging their blockers. Therefore machine output alone is not evidentiary warrant. $\square$

Audit 10.1 (AI-generated act attribution). *Claim: “AI output O proves that person P performed act A .” Target: act attribution. Carrier: AI system ledger, input data, validation, error profile, chain, and report standard. Active freedoms: model version, training contamination, input quality, base rate, threshold, false-positive rate, domain shift, interpretability, operator role, and human review. Laundering risks: output-to-proof, benchmark-to-case-reliability, authentication-to-correctness, and vendor-claim-to-carrier-adequacy. Possible final statuses: computation-only, authenticated machine output, evidence-compatible, method-valid within domain, probative-with-limits, or failed.*

11 Court-facing checklist and registry protocol

Legal claims are status-indexed not only by target and carrier, but by burden. Evidence sufficient for investigation may be insufficient for admissibility. Evidence sufficient for admissibility may be insufficient for proof. Evidence sufficient in a civil matter may be insufficient in a criminal matter.

Evidence sufficient to support an expert hypothesis may be insufficient to establish a legal element. Evidence sufficient to support a likelihood relation may be insufficient for categorical identity.

Protocol Result 11.1 (Burden laundering exclusion). *A claim supported under a lower or different burden standard cannot be reported as satisfying a higher or different legal burden unless the carrier discharges the additional status conditions required by that higher or different burden.*

Verification. A burden standard is part of the requested status class. If a claim satisfies only a lower or different burden, then the carrier conditions for the stronger status remain undischarged. Reporting the lower-burden claim in higher-burden language changes the requested status without changing the carrier. That is burden laundering. Therefore burden promotion requires discharge of the higher-status conditions. $\square$

Protocol Result 11.2 (Legal report preservation). *A fixed legal claim is not automatically reportable in the advocate’s preferred language. If the report changes target, carrier, scope, burden, uncertainty, permissible use, or nonclaim boundary, it creates a new claim requiring a new audit.*

Verification. A report is an uptake act. If the report omits a limitation, changes “consistent with” to “identified,” changes “admissible” to “proven,” changes “likelihood under population P ” to categorical certainty, changes “valid within domain D ” to universal validity, or changes “machine output” to “proof of the asserted legal target,” it changes the target or status of the claim delivered to the legal user. That new claim has not been audited unless the stronger report conditions have been satisfied. Therefore report preservation is required. $\square$

Table 18: Report-preserving legal language.

Overclaim risk	Preserved report
“Trace proves source”	“Trace is consistent with source under method M and limitations L .”
“Expert proved conclusion”	“Expert offered opinion C under method M , facts F , and limits L .”
“AI established fact”	“System S generated output O from input I under configuration K .”
“Witnesses independently confirm”	“Witnesses agree; independence status is X and contamination status is Y .”
“Document proves transaction”	“Document is authenticated for purpose P ; substantive truth requires carrier C .”
“Caused harm”	“Supports causal inference under carrier C , alternatives A , and standard S .”
“Admissible evidence proves element”	“Evidence is admissible for purpose P ; proof status remains separate.”

11.1 Court-facing warrant checklist

The equations are not the court-facing product. The practical product is a warrant checklist that can be used by a judge, lawyer, expert, forensic scientist, or reviewer without adopting AASC notation.

1. What exactly is being claimed?
2. What legal or evidentiary status is requested?
3. What jurisdiction, procedural posture, and burden apply?
4. What carrier supports that status?
5. What does the carrier actually establish, and what does it not establish?
6. What alternatives remain, and how are they legally disposed of?
7. What local promotion edge is being crossed, if any?
8. What wording would overstate the carrier?
9. What is preserved for the factfinder?
10. What is the strongest report-preserving formulation?

This checklist is the operational surface of the formal protocol. It helps prevent admissibility from being reported as proof, authentication as truth, expert qualification as method reliability, similarity as source identity, and machine output as proof of the asserted legal target.

11.2 Legal registry protocol

Protocol 11.1 (Legal AASC validator). *Every audited legal claim is processed through the following questions:*

1. *What legal/evidentiary claim is being asserted?*
2. *What exact legal target is requested?*
3. *What jurisdiction, procedural posture, and status standard apply?*
4. *What carrier purports to support the target?*
5. *Is the carrier adequate for the requested status?*
6. *What admissible rival hypotheses remain?*
7. *Which variations are harmless skin and which are content-changing?*
8. *Is there target, proxy, base-rate, method, machine, or report ancestry contamination?*
9. *Are evidence nodes independent and globally compatible?*
10. *Could status promotion be explained by laundering?*
11. *Does the report preserve the audited status?*
12. *What is the strongest defensible legal status?*
13. *What blocker prevents stronger status?*
14. *What next action would discharge the blocker?*

11.3 Registry schema

Table 19: Legal registry schema.

Field	Purpose
Claim ID	Stable identifier.
Literal claim	Exact assertion under audit.
Requested status	Legal status class asserted.
Jurisdiction/procedure	Forum, posture, rules, burden, permissible use.
Legal target	Object or status asserted.
Carrier	Evidence, method, authority, record, model, or ledger carrying warrant.
Carrier standard	Domain or jurisdiction standard required for status.
Evidence/provenance	Item identity, custody, source, preservation, authentication.
Method ledger	Validation, domain, error, application, limitations.
Alternatives	Rival hypotheses or legal characterizations.
Network audit	Independence, contamination, common ancestry, global compatibility.
Population/comparison	Base rate, comparison class, source class, reference population.
Machine ledger	System identity, version, data, validation, if applicable.
Active freedoms	Unresolved status-changing freedoms.
Closed freedoms	Freedoms discharged, declared, or legally excluded.
Laundering routes	Detected or excluded status-promotion routes.
Report object	Exact wording licensed by the audit.
Final status	Strongest defensible status.
Blockers	Active reasons stronger status is unavailable.
Next action	Minimal step to strengthen status.

Definition 11.1 (Legal audit certificate). *A legal audit certificate is*

$$\text{Cert}_J^\sigma(c) = (\text{Target}, \text{Carrier}, \text{Status}, \text{Blockers}, \text{Report}, \text{NextAction}).$$

The certificate does not certify truth, guilt, liability, or admissibility as a court ruling. It certifies whether the claim’s reported status is licensed by the declared legal audit basis.

11.4 Worked audit summary

11.5 Forensic trace

Table 20: Compressed forensic trace audit.

Field	Entry
Claim	Trace X identifies suspect Y .
Target	Source identity.

Field	Entry
Carrier	Chain ledger, validated method, comparison class, error model, alternative-source ledger, report standard.
Laundering risks	Similarity-to-identity; lab status-to-method; likelihood-to-certainty.
Likely status	Evidence-compatible, source-consistent, or likelihood-supporting unless stronger source-identification carrier is closed.
Report boundary	“Consistent with” or “likelihood under population P ” unless source-identifying status is discharged.

11.6 Expert testimony

Table 21: Compressed expert testimony audit.

Field	Entry
Claim	Expert E concludes C ; therefore C is established.
Target	Legal/factual conclusion C .
Carrier	Qualification, facts/data, method, application, fit, independence, report limits.
Laundering risks	Credentials-to-correctness; admissibility-to-proof; confidence-to-reliability.
Likely status	Expert-qualified-only, expert-opinion-admissible, probative-with-limits, or failed.
Report boundary	Opinion status must preserve method, facts, scope, and uncertainty.

11.7 Witness convergence

Table 22: Compressed witness convergence audit.

Field	Entry
Claim	Multiple witnesses independently corroborate event P .
Target	Independent convergence, not mere agreement.
Carrier	Witness observations plus independence and contamination ledger.
Laundering risks	Agreement-to-independence; common narrative-to-corroboration.
Likely status	Compatible-only, contaminated convergence, or independent corroboration if network closes.

Report boundary	Agreement and independence must be separately reported.
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11.8 Causation

Table 23: Compressed causation audit.

Field	Entry
Claim	Act A caused harm H .
Target	Jurisdiction-specific causation.
Carrier	Mechanism, expert analysis, alternative-cause ledger, standard of proof.
Laundrying risks	Sequence-to-causation; narrative-to-legal cause; correlation-to-causation.
Likely status	Sequence-compatible, causally-supportive, legally causal under standard, or failed.
Report boundary	Causal status must specify standard and alternatives.

11.9 AI or machine-generated evidence

Table 24: Compressed machine-evidence audit.

Field	Entry
Claim	Machine output O proves fact F .
Target	Fact attribution or classification.
Carrier	System identity, data provenance, validation, error profile, domain, chain, report.
Laundrying risks	Output-to-proof; benchmark-to-case-reliability; authentication-to-correctness.
Likely status	Computation-only, authenticated machine output, method-valid within domain, probative-with-limits, or failed.
Report boundary	Output must preserve system, input, domain, and validation status.

12 Legal proof as status-preserving construction

This section states the proof-theoretic contribution explicitly. The paper does not treat legal proof as the mechanical summation of evidence, the mere crossing of a numerical probability threshold, the selection of a persuasive narrative, or the announcement of an admissible evidentiary set. Those may be components of proof, and different legal systems may privilege them differently. The AASC claim is more basic: whatever proof theory a legal system uses, the final legal report must be a

status-preserving construction from a fixed target through an adequate carrier to a report whose wording remains inside the warrant supplied by that carrier.

The phrase *proof theory* is used here in a structural legal-epistemic sense: it denotes an account of the conditions under which a legal proof-report is warranted by a status-preserving construction. The paper does not claim to provide a sequent calculus, a formal derivation system, a normalization theorem, or a decision procedure for verdicts. Its claim is narrower: before a proof-report may be issued at a requested legal status, the construction from evidence to report must preserve the target, carrier, burden, factfinder allocation, and report boundary that make that status available.

Legal proof therefore has two layers. The first layer is domain-specific: rules of evidence, burdens, standards, factfinder roles, exclusionary policies, and local doctrines. The second layer is structural: the construction must not change target, carrier, alternative disposition, burden, factfinder allocation, or report status while presenting the result as the same claim. AASC/UEAP is a theory of the second layer. It does not replace the first layer; it prevents the first layer's outputs from being laundered into stronger reports.

Definition 12.1 (Proof-warrant construction). *For a claim c in jurisdictional ledger J at requested legal status σ , a proof-warrant construction is an ordered status-preserving path*

$$(T_J(c), C_J^\sigma(c), A_J^\sigma(c), B_J^\sigma(c), F_J(c), R_J^\sigma(c)),$$

where $T_J(c)$ is the fixed legal target, $C_J^\sigma(c)$ is the carrier offered for the requested status, $A_J^\sigma(c)$ records the disposition of admissible alternatives, $B_J^\sigma(c)$ records the burden or threshold condition, $F_J(c)$ records factfinder allocation and preserved roles, and $R_J^\sigma(c)$ is the final report object. The construction is status-preserving only if no step changes the asserted target or upgrades the report beyond the carrier, burden, alternative disposition, and factfinder role.

The proof-theoretic condition can be stated as:

$$\begin{aligned} & \text{ProofWarrant}_J^\sigma(c) \\ & \iff \\ & \text{ClaimStatusWarrant}_J^\sigma(c) \wedge \text{BurdenFit}_J^\sigma(c) \wedge \\ & \text{FactfinderPreserve}_J(c) \wedge \text{ReportBound}_J^\sigma(c). \end{aligned} \tag{2}$$

Here $\text{ClaimStatusWarrant}_J^\sigma(c)$ supplies target fixation, carrier adequacy, alternative disposition, no-laundering, and report preservation. $\text{BurdenFit}_J^\sigma(c)$ requires the construction either to satisfy the burden associated with σ or to report that it does not. $\text{FactfinderPreserve}_J(c)$ requires the construction not to usurp weight, credibility, or ultimate inference where J assigns those roles to the factfinder. $\text{ReportBound}_J^\sigma(c)$ requires the final statement to remain within the status actually constructed. The biconditional is internal to the declared audit basis. It is not a claim that every legal system defines proof in this way; it states what must hold for a legal proof-report to receive AASC/UEAP proof-warrant at the requested status.

Protocol Result 12.1 (Legal proof as status-preserving construction). *Under a declared legal audit basis, a legal proof-report at status σ is warranted only if the construction from evidence to report preserves target, carrier, burden, alternative disposition, factfinder allocation, and report limits. A probability, narrative, explanation, admissibility ruling, expert opinion, or machine output*

may function as a carrier within the construction, but it does not by itself license a stronger report unless the local promotion edge to that report is discharged.

Verification. A legal proof-report asserts more than the existence of evidence. It asserts that evidence, interpreted under a legal standard, supports a claim at a requested status. If the target changes, the report is about a different claim. If the carrier changes, the reported support is not the support offered. If alternatives remain in a report-limiting or fatal disposition, a stronger report overstates the construction. If the burden is not met, burden-satisfaction language is false to the ledger. If the factfinder role is bypassed, the report converts a preserved inference into an asserted conclusion. If the report exceeds the construction, the output is a new unaudited claim. Therefore a legal proof-report is warranted only when those coordinates are preserved. $\square$

This result explains why AASC/UEAP is not merely a checklist for evidence handling. It is a proof theory of lawful claim-status transport. Bayesian, narrative, explanatory, and hybrid accounts may disagree about how factfinders should reason from evidence to conclusion; AASC asks what those accounts must preserve before their output may be reported as a legal claim. A Bayesian posterior cannot erase target drift. A coherent narrative cannot convert compatibility into causation without a causal carrier. An inference to the best explanation cannot convert source consistency into source identity without method and comparison-class warrant. An admissibility ruling cannot convert expert testimony into proof of the asserted target. The issue is not which inferential style wins. The issue is whether the selected inferential style preserves the status it reports.

The same point gives the paper its relation to juridical proof. Legal proof is institutionally bounded: burdens differ, admissibility rules can be non-epistemic, exclusion can be normatively required, and factfinders retain roles that a report must not preempt. AASC/UEAP therefore does not offer an all-purpose metaphysics of legal truth. It supplies the structural warrant condition for legal reports: the report must be reachable from the carrier without illicit promotion.

13 Legal scholarship and doctrinal interfaces

The previous section stated the proof-theoretic result. This section places that result against existing legal doctrine and legal epistemology. It does not restate evidence doctrine. It shows where existing legal standards enter the AASC warrant basis and why the paper's contribution is a status-preservation layer rather than a replacement theory of law. The same carrier/report distinction recurs across expert evidence, authentication, hearsay, prejudice/use limits, forensic reform, comparative expert doctrine, and machine-generated evidence. Each doctrine marks a place where a legal report may not exceed the carrier supplied by the governing law.

The protocol enters a mature literature on proof, inference, scientific evidence, and forensic reliability. It does not displace that literature. It gives that literature a common proof-theoretic interface: whatever account of proof, inference, or admissibility a system adopts, a reported claim still requires a fixed target, a carrier adequate for the requested status, a lawful disposition of alternatives, exclusion of status laundering, and report language that preserves the carrier's limits.

13.1 Relation to theories of juridical proof

Juridical proof scholarship has long emphasized that factfinding is not the mechanical application of probability to isolated propositions. Evidence must be organized, interpreted, tested against

alternatives, and placed within institutional burdens and procedural roles. Work on evidence and inference networks, inference to the best explanation, legal epistemology, and scientific evidence in court shows that legal proof depends on relevance, explanatory integration, burdens, factfinder allocation, and normative constraints (Twining, 2006; Schum, 1994; Pardo and Allen, 2008; Laudan, 2006; Haack, 2014). AASC does not choose among Bayesian, explanatory, narrative, or hybrid theories of proof. It supplies a structural warrant condition for all of them: before an inferential output may be reported as a legal claim at a requested status, the output must preserve target, carrier, burden, alternative disposition, factfinder role, and report boundary.

This location matters. Bayesian accounts can update degrees of belief while still requiring a target and carrier for the event, source, or legal element being updated. Narrative and explanatory accounts can compare competing stories while still requiring a carrier for the transition from coherence to causation or liability. Hybrid accounts can combine probability, explanation, and legal judgment while still needing report language that preserves the difference between admissibility, weight, sufficiency, and proof of the asserted legal target. AASC therefore functions as a proof theory of lawful claim-status transport. It does not say which inference theory a court must use. It says that no inference theory may silently promote a claim beyond the carrier it supplies.

13.2 Relation to forensic epistemology and expert evidence scholarship

Forensic evidence scholarship supplies the most direct external pressure on the protocol. The forensic reform literature has repeatedly warned against overstated source conclusions, weak validation, examiner subjectivity, categorical language, and confusion between method acceptance and case-specific accuracy (Mnookin et al., 2011; Kaye, 2010; Saks and Koehler, 2005; Cole, 2001). AASC re-expresses those concerns as target/carrier/report problems: the target may be source consistency rather than source identity; the carrier may support likelihood rather than categorical attribution; and the report may preserve or launder that distinction.

The same structure appears in expert evidence. A qualified expert is not the same thing as a reliable method. A reliable method in general is not the same thing as reliable application to the facts of the case. Reliable application to the case is not permission to state more than the method supports. These distinctions are familiar in doctrine, but AASC provides a general audit vocabulary for registering them across expert reports, forensic reports, machine outputs, and legal summaries. The result is neither anti-expert nor anti-forensic. It is anti-overstatement: the audit preserves the strongest status the carrier actually warrants and blocks only the unsupported promotion.

13.3 What AASC adds to existing doctrine

The paper's contribution is therefore narrow but structural. Evidence doctrine supplies admissibility rules, exclusionary policies, burdens, standards, and institutional roles. Proof theory supplies accounts of how factfinders may reason from evidence to conclusions. Forensic science supplies method-specific validation, error-rate, and reporting disciplines. AASC supplies the cross-status audit layer connecting them: it identifies the requested status, the carrier offered for that status, the local promotion edge being crossed, the alternatives that remain, and the report language that would preserve or launder the result.

The matrix below states the paper's relation to existing legal scholarship. Its purpose is not to displace any theory of juridical proof, but to identify the status-preservation layer that those

theories still require.

Table 25: Positioning matrix for legal evidence scholarship.

Literature	Primary concern	AASC contribution
Evidence doctrine	Admissibility, permissible use, exclusion, and procedural gates.	Status-promotion audit across doctrine-supplied standards.
Proof theory	Standards of proof, juridical inference, narrative/probability/explanation.	Carrier/report separation and local edge control.
Forensic epistemology	Reliability, source attribution, method validation, and reporting limits.	Similarity-to-identity and likelihood-to-certainty firewall.
Expert evidence	Qualification, method, application, independence, and fit.	Authority, method, application, and report decomposition.
Legal epistemology	Knowledge, justification, error allocation, institutional factfinding.	Claim-status warrant grammar under declared legal roles.
AI evidence	Machine reliability, explainability, authentication, and domain shift.	Computation-laundering audit with system/data/error/domain ledger.

The central legal claim is therefore precise rather than merely modest. AASC does not replace Bayesian, narrative, explanatory, or doctrinal accounts of proof. It supplies the proof-theoretic transport rule that those accounts still need: a proof theory, evidentiary rule, forensic method, expert report, or machine output cannot be reported at a status its own carrier does not warrant.

13.4 Doctrinal interfaces: where AASC enters and where it must stop

The protocol is legally useful only if it respects doctrine. AASC does not supply the rule of decision. It supplies a status-preservation map showing where a doctrine-supplied standard enters the audit basis and which report transitions remain unlicensed.

13.5 Rule 702, Daubert, and Kumho as carrier-standard inputs

Federal Rule of Evidence 702 is naturally represented as an expert-carrier standard. It separates qualification, helpfulness, sufficient facts or data, reliable principles and methods, and reliable application to the facts. The 2023 amendment and committee materials emphasize that the proponent must demonstrate the Rule 702 requirements by the Rule 104(a) preponderance standard and that an expert opinion must not overstate what follows from reliable application of the basis and methodology (Fed. R. Evid. 702, 2026). Daubert supplies the reliability/relevance gatekeeping

frame for scientific expert testimony, and *Kumho* extends gatekeeping across technical and other specialized expert knowledge (*Daubert*, 1993; *Kumho Tire*, 1999).

AASC does not duplicate those rules. It uses them as the declared C_j^g carrier standard when the requested status is expert-opinion admissibility or probative expert assistance. The audit then prevents credentials from being laundered into method reliability, general method validity into reliable case application, and admissibility into proof.

13.6 Case-grounded demonstration: *Daubert/Kumho* as carrier and report-boundary cases

The *Daubert/Kumho* line is useful because it shows the protocol recovering familiar legal distinctions rather than floating above doctrine. In *Daubert*, the gatekeeping problem concerns whether the offered scientific expert testimony has a reliable methodology and fits the legal target for which it is offered. In *Kumho Tire*, the same gatekeeping discipline extends beyond conventional scientific testimony to technical and specialized expert knowledge. Together, the cases show that expert qualification, general method acceptability, case-specific application, and final report language are distinct coordinates.

In AASC terms, qualification is a role coordinate; method reliability is a carrier coordinate; application is a case-specific carrier coordinate; fit is target alignment; and the final opinion is the report object. A *Daubert/Kumho*-style exclusion, limitation, or narrowing is therefore not an AASC novelty. It is a doctrinally familiar instance of claim-status warrant control: the court prevents an expert report from crossing a local promotion edge without the carrier required for that edge. The admissibility of an expert opinion is not proof of the asserted target; general method reliability is not reliable application here; and reliable application here is not permission to state more than the method can support.

13.7 Rule 403 as report/use and prejudice-risk boundary

Rule 403 is not merely an evidence-volume rule. In AASC terms it is a report/use constraint. Even evidence with target alignment and carrier adequacy may be excluded or limited if its probative value is substantially outweighed by unfair prejudice, confusion, misleading the jury, delay, waste, or cumulative presentation (Fed. R. Evid. 403, 2026). That means the status vector must distinguish epistemic/probative support from admissibility and permissible use. A claim can be epistemically probative yet legally excluded or limited.

13.8 Rule 901 as authentication target, not truth target

Rule 901-type authentication is a target-specific carrier: the proponent must produce evidence sufficient to support a finding that the item is what the proponent claims it is (Fed. R. Evid. 901, 2026). Authentication does not by itself establish the truth of a document's contents, the reliability of a machine process, the source identity of a trace, or the legal effect of an item. The AASC audit therefore treats authentication as its own target coordinate and blocks authentication-to-truth laundering.

13.9 Hearsay, proxy carriers, and non-epistemic exclusions

Hearsay doctrine shows why legal admissibility is not reducible to raw epistemic relevance. A statement may be relevant and meaningful while its carrier is restricted because the declarant is absent, cross-examination is unavailable, or the statement is offered for a truth purpose. In AASC terms, the statement may have a semantic target but lack the legally required carrier for the truth-status asserted. Privilege, confrontation, constitutional exclusion, policy exclusions, and sanctions reinforce the point: legal admissibility contains institutional and normative coordinates in addition to epistemic support.

13.10 Mohan and White Burgess as comparative carrier standards

Canadian expert-evidence doctrine supplies a useful comparative ledger. Mohan states the classic admissibility criteria: relevance, necessity in assisting the trier of fact, absence of an exclusionary rule, and a properly qualified expert (*Mohan*, 1994). White Burgess adds the expert’s duty to provide fair, objective, and non-partisan assistance to the court (*White Burgess*, 2015). In the AASC registry, these become carrier, independence, and reportability fields. A qualified expert can still fail independence, method, application, or report-preservation coordinates.

13.11 Forensic reform as method-carrier pressure

The National Academies report and PCAST report are best read here as carrier-adequacy pressure points, not as automatic admissibility rules (National Research Council, 2009; President’s Council of Advisors on Science and Technology, 2016). They identify why source-identification claims require method validation, standards, error-rate discipline, proficiency, comparison-class control, and report-language limits. Under PCAST’s proposed foundational-validity standard, certain feature-comparison disciplines were judged not to have met that standard at the time of review; that should not be over-reported as a universal legal inadmissibility conclusion. AASC’s role is narrower: if a forensic claim requests source-identifying status, the carrier must include the discipline-specific validation and reporting conditions required by the declared legal basis.

Table 26: Illustrative forensic carrier contrasts.

Discipline type	Typical target	Carrier pressure	AASC overclaim risk
Single-source DNA	Source inclusion or exclusion under a genetic profile	Laboratory protocol, contamination controls, population genetics, statistical reporting	Match or likelihood language reported as categorical certainty.

Discipline type	Typical target	Carrier pressure	AASC overclaim risk
Latent fingerprint comparison	Source consistency or identification under discipline-specific criteria	Image quality, feature sufficiency, examiner discretion, verification, error-rate evidence	Examiner confidence or historical acceptance reported as complete source certainty.
Firearms/toolmarks	Cartridge or toolmark compatibility, likelihood, or source attribution	Validation studies, subclass characteristics, comparison class, examiner discretion, error rates, report language	Similarity or “identification” terminology reported beyond method validation.
Bitemark comparison	Association or exclusion, if any, under severe method limits	Biological distortion, uniqueness assumptions, method validation, error rates	Pattern resemblance reported as source identity.

13.12 Machine-generated evidence and proposed Rule 707

Machine-generated evidence exposes computation laundering. Proposed Rule 707 is treated here only as pending rulemaking evidence that machine-generated evidence has become a recognized reliability problem; the argument does not depend on adoption of the rule or on any final wording. The Advisory Committee’s Rule 707 materials identify a possible gap where machine output is offered without an expert witness even though the same conclusion would trigger Rule 702 if offered by a human expert (Advisory Committee on Evidence Rules, 2026). At the time of writing, procedural-status reporting indicates that the proposal remains unsettled and has been delayed for further discussion (Raymond, 2026). The present paper therefore treats proposed Rule 707 only as illustrative rulemaking material.² The protocol does not depend on adoption of that rule. Its AASC point is stable: machine output requires system identity, input provenance, validation, error profile, domain limits, and report-preserving language before it can support more than computation-only or authenticated-output status.

14 Objections and replies

14.1 This is just evidence law in new vocabulary

The protocol does not replace evidence law. It records how evidence-law standards enter a status audit. Its contribution is cross-status discipline: admissibility is not proof, expert authority is not

²Proposed Rule 707 is cited only as rulemaking evidence that machine-generated evidence has become a recognized evidentiary reliability problem. The protocol does not depend on adoption of the proposed rule, and the paper makes no claim about its final content, timing, or enactment.

method warrant, similarity is not source identity, and computation is not proof of the asserted legal target.

14.2 This invades the role of judge or jury

The protocol does not decide admissibility, weight, credibility, guilt, liability, or remedy. It produces a registry row showing what status the claim is licensed to report under a declared audit basis. A judge, jury, agency, or tribunal retains the legal role assigned by the jurisdiction.

14.3 Cross-examination already handles this

Cross-examination can expose blockers. AASC identifies where those blockers live: target, carrier, chain, method, population, alternatives, evidence network, laundering route, burden, or report object. It prevents the blocker from being lost when the claim is summarized downstream.

14.4 Legal standards are flexible, not formal

AASC does not require all legal standards to be rigid. It requires the requested status and carrier standard to be declared. Flexible standards can still be audited: the audit records the flexibility as part of J and prevents a claim from being reported as stronger than that standard permits.

14.5 Probative force is holistic

Holistic evidence can be represented as a network. The protocol denies only illicit amplification, not ordinary legal weighing. A holistic network may support a strong status if it is target-aligned, sufficiently independent, globally compatible, and adequate under the jurisdictional standard.

14.6 Authority matters in law

Authority matters when authority is the legal target: a statute, rule, precedent, order, certification, or institutional act. Authority does not substitute for factual, causal, forensic, or methodological warrant when those are the targets.

14.7 The paper smuggles in AASC

The kernel explains why the protocol tracks target, carrier, admissibility, and irreversibility. The legal audit results are conditional protocol results. A reader may reject the kernel and still use the registry operationally as a status-preservation tool, provided the audit basis is declared.

15 Conclusion

Legal inference is full of useful lower-status claims. A trace may be consistent with a source. An expert may be qualified. A document may be authenticated. A method may be valid within a domain. A machine may generate an output. Witnesses may agree. A narrative may be coherent. Evidence may be admissible. Each status can be valuable. The failure occurs when one status is reported as another without discharging the blocker between them.

AASC/UEAP gives law a compact audit rule:

No legal or evidentiary claim should be reported at a status stronger than its target, carrier, jurisdictional standard, alternative disposition, laundering audit, and report object permit.

The legal audit developed here preserves compatibility as compatibility, admissibility as admissibility, expertise as expert assistance, similarity as similarity, likelihood as likelihood, machine output as machine output, and causation as causation only when the carrier for causation is fixed. It is not a replacement for law. It is a report-preserving status firewall for legal claims.

A legal system can tolerate uncertainty, competing narratives, flexible standards, and factfinder judgment. It cannot safely tolerate undisclosed status promotion. A claim may be admitted, weighed, argued, or preserved for the factfinder only under language that remains faithful to the carrier that actually supports it.

A Legal audit-result dependency map

Table 27: Legal audit-result dependency map.

Legal result	AASC/UEAP dependency	Blocker closed
Legal Target-Fixation Necessity	Reference / target fixation	target drift
Carrier Adequacy Is Status-Relative	Standing / carrier adequacy	wrong support
Legal Freedom Disposition	Admissibility / alternative disposition	hidden rival or factfinder-preserved alternative
Legal Laundering Exclusion	Irreversibility / no repair	status promotion
Legal Evidence Non-Amplification	Evidence network / alternative disposition	false independence
Similarity Is Not Source Identity	Target/carrier alignment	source/proxy laundering
Machine Output Does Not Supply Evidentiary Warrant	Computation laundering	computation-only
Burden Laundering Exclusion	Requested status class	burden mismatch
Legal Report Preservation	Report object	over-reporting

B Blank legal registry template

Field	Entry
Claim ID	
Literal claim	
Requested status	

Jurisdiction/procedure
 Legal target
 Carrier
 Carrier standard
 Evidence/provenance
 Method ledger
 Alternatives
 Network audit
 Population/comparison
 Machine ledger
 Active freedoms
 Closed freedoms
 Laundering routes
 Report object
 Final status
 Blockers
 Next action

C Completed registry rows

Table 29: Compressed completed rows.

Claim ID	Claim text	Final status	Report boundary
LEGAL-FOR-001	Trace X identifies suspect Y .	Evidence-compatible or source-consistent unless full source-identification carrier is closed.	Do not report similarity as identity.
LEGAL-EXP-001	Expert E concludes C ; therefore C is established.	Expert-qualified-only, admissible expert opinion, or probative-with-limits.	Do not report expert status as proof.
LEGAL-WIT-001	Multiple witnesses independently corroborate event P .	Compatible-only unless independence ledger is fixed.	Report agreement separately from independence.
LEGAL-CAU-001	Act A caused harm H .	Sequence-compatible, causally-supportive, or causal under standard.	Preserve causal standard and alternatives.

Claim ID	Claim text	Final status	Report boundary
LEGAL-AI-001	Machine output O proves fact F .	Computation-only or method-valid within domain unless validation carrier closes.	Preserve system, input, validation, and domain.

D Expanded legal laundering table

Route	Typical signal	Possible status
Authority	Conclusion is accepted because of title, lab, institution, or prior admissibility.	authority-only or carrier-unfixed
Method	Named method is used without validation or application record.	method-unfixed or domain-dependent
Chain	Evidence continuity is assumed or compressed.	chain-unfixed or authenticated-only
Similarity	Observed feature agreement becomes identity.	evidence-compatible or source-consistent
Narrative	Coherent reconstruction is treated as proof.	compatible-only
Witness network	Agreement is counted without independence audit.	evidence-compatible
Baseline	Population or base rate is hidden.	baseline-blocked
Burden	Lower threshold is reported as higher threshold.	status-class mismatch
Admissibility	Received evidence is described as established fact.	report failure
Machine	Output is treated as reliable legal fact.	computation-only
Authentication	Item identity is treated as truth of contents.	authenticated-only
Causation	Sequence or correlation is treated as legal cause.	causally-unfixed

E Report-language crosswalk

Unsafe report	Safer report-preserving formulation
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“ <i>X</i> proves <i>Y</i> .”	“ <i>X</i> supports <i>Y</i> at status <i>s</i> under carrier <i>C</i> and limitations <i>L</i> .”
“The expert proved causation.”	“The expert offered a causal opinion under method <i>M</i> , facts <i>F</i> , and standard <i>S</i> .”
“The AI found the truth.”	“System <i>S</i> generated output <i>O</i> under configuration <i>K</i> from input <i>I</i> .”
“Witnesses independently agree.”	“Witnesses agree; independence status is [fixed/unfixed/contaminated].”
“The document proves the transaction.”	“The document is authenticated; substantive transaction proof requires additional carrier <i>C</i> .”
“The trace came from the suspect.”	“The trace is consistent with the suspect under method <i>M</i> and report boundary <i>R</i> .”

F Source hierarchy and update discipline

Legal citation hygiene matters for a cross-domain audit paper. The manuscript uses sources in the following order of authority: enacted rules, official rule text, and official advisory committee materials first; official court decisions next; government or court-published reports and standards next; peer-reviewed or scholarly commentary next; news and legal-industry reporting only for procedural status or current rulemaking context. Federal Rules of Evidence 702, 901, and 403 are treated as rules; Daubert, Kumho, Mohan, and White Burgess are treated as cases; Advisory Committee materials are treated as rulemaking materials; and Reuters-style news sources are used only for contemporary procedural status. Proposed Rule 707 and related machine-generated-evidence materials are cited as rulemaking materials, not operative doctrine. Before jurisdiction-specific filing, submission, or legal use, all rule text, proposed-rule status, and local doctrine must be updated against official sources.

G U.S./Canada doctrinal interface note

This appendix records how the paper uses U.S. and Canadian sources. Rule 702, Daubert, Kumho, Rule 901, and Rule 403 are used as examples of U.S. federal carrier and report constraints, not as universal law (United States Courts, 2024; Fed. R. Evid. 702, 2026; Fed. R. Evid. 901, 2026; Fed. R. Evid. 403, 2026; *Daubert*, 1993; *Kumho Tire*, 1999). Mohan and White Burgess are used as Canadian expert-evidence anchors (*Mohan*, 1994; *White Burgess*, 2015). Nothing in this paper asserts that a court must apply AASC or that an AASC certificate has legal force. The protocol is an audit discipline for document review, research, and status-preserving analysis.

H AI/machine-generated evidence ledger

Ledger field	Required content
System identity	Vendor, model, version, checkpoint, configuration, software/hardware environment.

Input ledger	Data source, quality, chain, preprocessing, operator inputs, prompts, thresholds.
Training/exposure	Known training data, benchmark exposure, leakage risks, domain tuning.
Validation	Test domain, accuracy, calibration, robustness, error rate, external validation.
Uncertainty	Confidence meaning, false positives, false negatives, base-rate sensitivity.
Bias/domain shift	Population coverage, subgroup effects, operational context.
Human role	Operator, reviewer, override, interpretation, and accountability.
Reproducibility	Logs, seeds if applicable, model determinism, output preservation.
Report boundary	What the output may be called: generated classification, machine opinion, method-valid output, or proof of the asserted legal target.

I Nonclaim boundaries

This manuscript is a research article in legal audit-protocol development. It is not legal advice, not a litigation strategy, not an expert report, not an admissibility motion, not a judicial opinion, not a forensic validation study, and not a certification of any real evidence item. A legal user must supply jurisdiction-specific doctrine, facts, evidence, and professional judgment. AASC can preserve status; it cannot lawfully replace the institutions that assign legal effect.

Declarations and source status

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Source status. The paper is built as a legal-domain application of the UEAP claim-standing source manuscript (Maley, 2026a). External legal authorities are used only as domain-standard anchors and should be updated before filing or submission in any jurisdiction-specific venue. Proposed Rule 707 is treated as unsettled rulemaking material and not as operative doctrine.

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